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Positioning accuracy prediction of precision cycloid reducer based on a BDDTE model and use in design optimization

Xincheng Wang [a], Luyang Li [b], Linbo Hao [a], Huaming Wang [a,*]

[a] College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, 29 Yudao Street, Nanjing, 210016, China
[b] College of Mechanical Engineering, Yangzhou University, 196 West Huayang Street, Yangzhou, 225127, China

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ABSTRACT

As users' expectations for precision cycloid reducer positioning accuracy rise, accurate positioning accuracy prediction of products before assembly and design solutions to improve product qualification rates are very important. Considering the actual manufacturing conditions of components, a new multi-source error (MSE) equivalence modeling assumption and pin gear's MSE measurement method are proposed, and the rationality of the assumption is verified through experiments. Based on the geometric characteristics of each component's MSEs, a new analytical model of bi-directional drive transmission error (BDDTE) is established, which organically unifies the prediction of transmission error (TE) and lost motion (LM). The influence of the output mechanism on the positioning accuracy of the entire reducer is found to be non-negligible through Monte Carlo simulation; meanwhile, the general tolerance allocation scheme makes it difficult to ensure that each prototype meets the performance requirements. For this reason, a new design solution for the positioning accuracy of the reducer with the pin gear and output mechanism as the optimization targets is proposed. Finally, the prototype test results show that the BDDTE model based on the new assumption has a higher accuracy in predicting positioning accuracy compared with the model based on the conventional assumptions. Meanwhile, the effectiveness of the proposed improvement design solution for positioning accuracy improvement is verified. The BDDTE model reveals the intrinsic connection between the TE and LM of the cycloid reducer, and the improvement solution has guiding effects on improving the product qualification rate.

1. Introduction

Unloaded transmission error (TE) and lost motion (LM) are key performance indicators [1] for the evaluation of the positioning accuracy of precision cycloid reducers [2]. As the demand for precision cycloid reducer products in industrial robots, aerospace, medical equipment, and other fields of application [3] is increasing, users' requirements for positioning accuracy are also increasing. Tremendous investigations have been carried out on the single-stage cycloid drive, including transmission principles analysis [4–10], tooth contact analysis [11–15], cycloid tooth profile modification design [16–19], manufacturing errors modeling [20–24], and dynamic analysis [25–29].

The unloaded TE, as a key performance index for the evaluation of the positioning accuracy of precision cycloid reducers, is one of the main objects of research by scholars. Sun et al. [30] carried out a numerical simulation of the backlash of the RV reducer based on the Monte Carlo method, which helps predict the positioning performance of the RV reducer. Lin et al. [31] designed a new two-stage cycloid gear reducer based on topological analysis and carried out the modification design of cycloid tooth profile, after which the reducer kinematic errors were analyzed by TCA. Li et al. [32] proposed a loaded contact analysis model based on the TCA method and studied the influence of tooth side clearance on the contact force and TE of the cycloid pin gear transmission pair. Ren et al. [33] proposed a tooth profile modification method based on five reference points and established a nonlinear dynamic model for analyzing the TE of the cycloid drive. Xu et al. [34] considered the influence of friction torque on the unloaded TE test of the reducer and proposed an optimal motor speed calibration method to achieve an accurate measurement of the unloaded TE. Li et al. [35] proposed a tooth profile modification method aiming at pressure angle optimization, and the test results of a prototype verify the effectiveness of this design method for the TE control. Yang et al. [36] established an unloaded TE model for the RV reducer based on the incremental loop method, substituted the components' error measurement results into the

* Corresponding author.
E-mail addresses: wxc787@nuaa.edu.cn (X. Wang), llyyzu123@163.com (L. Li), lb_hao@nuaa.edu.com (L. Hao), hmwang@nuaa.edu.cn (H. Wang).

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model to predict the performance of the reducer, and verified the model's effectiveness through prototype testing. Xu et al. [37] designed a cycloid reducer with a rotatable output-pin mechanism and established a dynamic model considering manufacturing errors, and verified the effectiveness of the model through tests. Tariq et al. [38] conducted an evaluation and comparison between a conventional cycloidal drive and its involute-tooth kinematic equivalent using finite element analysis and dynamic multi-body analysis, revealing that the involute-tooth design is more efficient. Kalligeros et al. [39] simulated double-flank inspection for gears with diverse profile deviations, yielding results consistent with existing theoretical and experimental research. Wang et al. [40] introduced a novel theoretical approach for compensating cycloidal gear machining errors by proposing a Monte Carlo simulation-based model that accounts for the random nature and diverse distribution patterns of machining errors.

The lost motion [41], as another key performance index for precision cycloid reducer positioning accuracy evaluation, has not received much attention from scholars for a long time, and the development of theoretical analysis models has been slow. Sun et al. [42] established an LM model of a cycloid reducer considering the influence of manufacturing errors and tooth profile modification. The validity of the analysis result was verified through tests, and the test results showed that the LM values of 30 different output angle positions of the reducer were not equal and deviated significantly. Similarly, Xu et al. [43] and Li et al. [44] found that the cycloidal reducer geometric return value varies significantly with the rotational position of the output by testing. The conventional LM analysis model does not sufficiently consider the global characteristics of the cycloid reducer. However, progress has been made in the analysis of the return difference of planetary reducers. Zhang et al. [45, 46] established a bi-directional transmission model to adequately predict the planetary gearbox's LM and optimized the LM based on this model. Therefore, a complete analysis model is needed for the accurate evaluation of the LM at different angular positions of the output of the entire cycloid reducer.

The manufacturing error and assembly error, as the key input parameters for the positioning accuracy analysis of the reducer, directly affect the accuracy of the analysis results. Lin et al. [47] conducted simulations of the unloaded TE and backlash for 2000 single-stage cycloid drives based on the Monte Carlo method and optimized the reducer tolerance allocation scheme, but the maximum value of TEs for the optimized 2000 virtual prototypes was more than 120 arcsecs, and the minimum value of backlash was more than 300 arcsecs. Wu et al. [48] carried out a numerical simulation of the tolerance design based on the Monte-Carlo method and studied the sensitivity of various influencing factors to the TE of the RV reducer. Jiang et al. [49] analyzed the statistical distribution of 10,000 virtual prototypes unloaded TEs based on the equivalent dynamic model. The results show that the deviation between the best and worst TE performance in the virtual prototype exceeds 100 arcsecs.

To sum up, the following problems exist in the current research on the positioning accuracy of cycloid reducers.

(1) The theoretical research is still focused on the single-stage cycloid drive. The theoretical research on the output mechanism, especially the pin-type output mechanism, is still relatively small and imperfect. The conventional TE analysis model ignores the difference between the forward drive TE and the reverse drive TE of the reducer. The conventional LM analysis model cannot evaluate the global performance of the reducer at different output angular positions. Thus, there is a lack of a unified model that can comprehensively evaluate the positioning accuracy of the entire reducer.
(2) Different equivalence modeling assumptions of the MSEs may have an impact on the results of the positioning accuracy prediction. Thus, an equivalent modeling method for geometric features of MSEs based on actual manufacturing process analysis is needed, and its rationality should be verified by experiments.
(3) The deviation in positioning accuracy of the prototypes manufactured based on the same tolerance allocation scheme may be large, and it is difficult to ensure that all the prototypes meet the performance requirements of the precision reducer with the general tolerance allocation optimization. Thus, a more applicable solution for improving the positioning accuracy of the reducer is needed.

This study aims to establish a positioning accuracy prediction model for the cycloid reducer and to further study the performance improvement design method. In Section 2, the geometric characteristics of multi-source errors (MSEs) are modeled; In Section 3, a bi-directional drive transmission error (BDDTE) model of the entire reducer is established; In Section 4, the equivalent modeling assumptions of the pin gear MSEs are investigated, the virtual prototype Monte Carlo simulation is carried out, and a reducer positioning accuracy improvement solution is proposed.; In Section 5, the pin gear MSEs measurement method is presented, the positioning accuracy of the unassembled prototype is predicted and improved, and the positioning accuracy tests are conducted on the prototype before and after the improvement, respectively.

2. Cycloid reducer multi-source errors analysis and equivalent modeling

2.1. Cycloid reducer design

The cycloid drive studied in this paper is the structure with one tooth difference, and the tooth profile of the cycloid gear is a kind of epicycloid curve [50]. The standard tooth profile [51] is designed by Eq. (1). In Eq. (1), Z_c is the number of cycloid gear teeth, Z_p is the number of pin gear teeth, $i^H = Z_p/Z_c$. R_p is the radius of the pin. R_p is the pin distribution radius of the pin gear. K_1 is the short-amplitude coefficient, $K_1 = eZ_p/R_p$. E is the input crankshaft eccentricity. Define $S_c = 1 + K_1^2 - 2K_1 \cos\varphi_c$, and φ_c is the angular parameter.

$$\begin{cases} x_c = -\left(R_p - r_p S_c^{-\frac{1}{2}}\right)\sin\left[\left(1-i^H\right)\varphi_c\right] - \left(e - K_1 r_p S_c^{-\frac{1}{2}}\right)\sin\left(i^H\varphi_c\right) \\ y_c = \left(R_p - r_p S_c^{-\frac{1}{2}}\right)\cos\left[\left(1-i^H\right)\varphi_c\right] - \left(e - K_1 r_p S_c^{-\frac{1}{2}}\right)\cos\left(i^H\varphi_c\right) \end{cases} \tag{1}$$

In Fig. 1, the ideal cycloid drive has all pin teeth in contact with the cycloid gear and half of the teeth involved in carrying the load at the same time. P is the intersection of the normals of all meshing points. α_i is the included angle between O_pO_c and O_pO_i, which can be expressed as Eq. (2). In ΔO_iPO_p, Eq. (3) can be deduced from the sine theorem. ε_i is the angle between the normal direction of the meshing point of the i-th pin and the Y_c axis, and ε_i can be expressed as Eq. (4). Allow the number of the closest cycloid gear tooth and pin in the counterclockwise direction of the rotating arm O_PO_C to be 1, and the number rises counterclockwise. The number of cycloid gear teeth that contact the i-th pin is j.

$$\alpha_i = \mathrm{mod}\left(\varphi_{in}, \frac{2\pi}{Z_p}\right) + \frac{2\pi(i-1)}{Z_p} \tag{2}$$

$$\sin\beta_i = \frac{\sin\alpha_i}{\sqrt{1 - 2K_1\cos\alpha_i + K_1^2}} \tag{3}$$

$$\varepsilon_i = \pi - \beta_i - \varphi_{in} - \varphi_{out} \tag{4}$$

In Fig. 2, a cycloid reducer with a pin shaft output mechanism is designed. The cycloid gear and the input crankshaft are supported by the bearing. The pins are installed in the tooth grooves of the case and

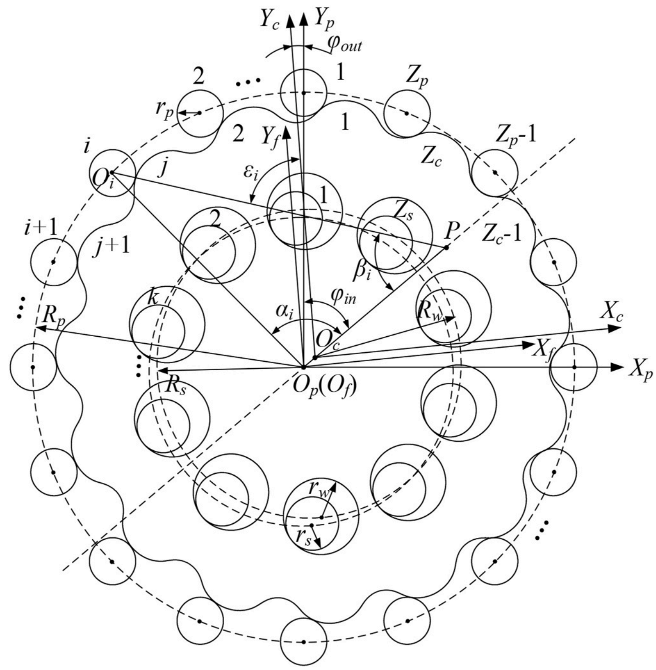

Fig. 1. Schematic diagram of the ideal cycloid drive.

combined into the pin gear. An output pin shaft is fixedly installed in the output flange, and a pin shaft sleeve is installed outside the pin shaft. Each pin sleeve cooperates with the pin hole of the cycloid gear to transmit the revolution of the cycloid gear to the output flange. Among them, the main design parameters of the reducer include that the number of cycloid gear teeth Z_c is 51, the number of pin gear teeth Z_p is 52, the pins distribution radius R_p is 30 mm, the pin radius is 0.975 mm, the crankshaft eccentricity e is 0.45 mm, the number of the output pin shaft Z_s is 10, the pin hole radius R_w is 4.5 mm, and the pin sleeve radius R_s is 4.05 mm.

Due to unavoidable manufacturing and assembly errors, the cycloid gear tooth profile should be modified to actively generate the tooth-side clearance so as to meet the assembly conditions. The isometric and offset combination modification [16,42,52] are adopted for the cycloid tooth profile modification, and the modified tooth profile is expressed as Eq. (5). ΔR_p is the isometric modification parameter, and Δr_p is the offset modification parameter. Define $S_{cm} = 1 + K_1'^2 - 2K_1' \cos\varphi_c$, and $K_1' = eZ_p/(R_p + \Delta R_p)$.

$$\begin{cases} x_{cd} = -\left[R_p + \Delta R_p - (r_p + \Delta r_p) S_{cm}^{-\frac{1}{2}}\right] \sin\left[\left(1 - i^H\right)\varphi_c\right] - \left(e - K_1' r_p S_{cm}^{-\frac{1}{2}}\right) \sin\left(i^H \varphi_c\right) \\ y_{cd} = \left[R_p + \Delta R_p - (r_p + \Delta r_p) S_{cm}^{-\frac{1}{2}}\right] \cos\left[\left(1 - i^H\right)\varphi_c\right] - \left(e - K_1' r_p S_{cm}^{-\frac{1}{2}}\right) \cos\left(i^H \varphi_c\right) \end{cases} \tag{5}$$

The values of MSEs are uncertain in the theoretical design stage and usually need to be equivalently modeled based on the probability density sampling method. The main components of the reducer are manufactured using precision grinding [53], and the manufacturing accuracy can reach the standard of IT4 to IT5. The cycloid gear center hole eccentricity error is related to the positioning accuracy of the workpiece clamping, which can be controlled by precision measurement to less than 2 μm. The tolerance allocation scheme is shown in Table 1. Since there is a certain deviation between the actual manufactured cycloid tooth profile and the design result, it is necessary to select the modification amounts to dress the grinding wheel and manufacture the cycloid gear first and model the tooth profile error equivalently based on the measured results. The cycloid tooth profile modification amounts are $\Delta r_p = -6$ μm and $\Delta R_p = -10$ μm, respectively.

2.2. Positioning accuracy indicators

Fig. 3 shows the schematic diagram of the bi-directional drive transmission error (BDDTE) of the entire reducer, and the positioning accuracy of the prototype should be evaluated by several performance

Table 1
Tolerance allocation scheme.

Parameters	Tolerance
Pin distribution radius error δ_{Rp}	[5,10] μm
Cycloid gear tooth profile error δ_{cp}	[-5,5] μm
Pin gear tooth groove profile error δ_{rg}	[-5,5] μm
Pin radius error δ_{rp}	[-1.5,0] μm
Cycloid gear tooth dividing angle error λ_c	[-15,15] arcsecs
Pin gear tooth dividing angle error λ_p	[-15,15] arcsecs
Crankshaft eccentricity error δ_e	[-4,4] μm
Cycloid gear center hole eccentricity error δ_{cr}	[0,2] μm
Output flange pin distribution radius error δ_{Rs}	[-2,2] μm
Cycloid gear pin hole distribution radius error δ_{Rw}	[-2,2] μm
Output pin sleeve radius error δ_{rs}	[-2,0] μm
Output pin hole radius error δ_{rw}	[3,5] μm
Output pin shaft dividing angle error λ_s	[-15,15] arcsecs
Cycloid gear pin hole dividing angle error λ_w	[-15,15] arcsecs

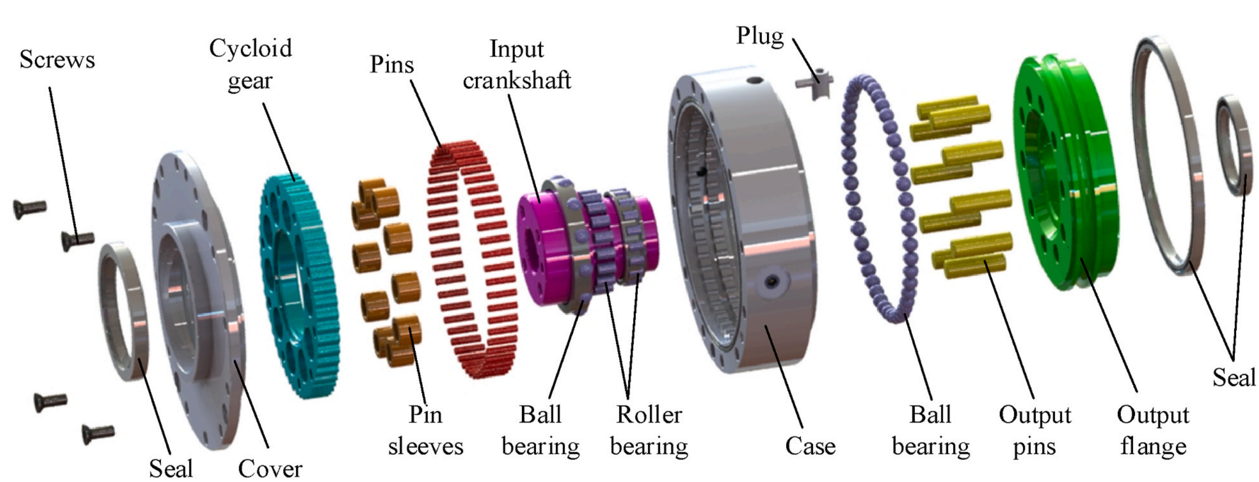

Fig. 2. Exploded diagram of the cycloid reducer.

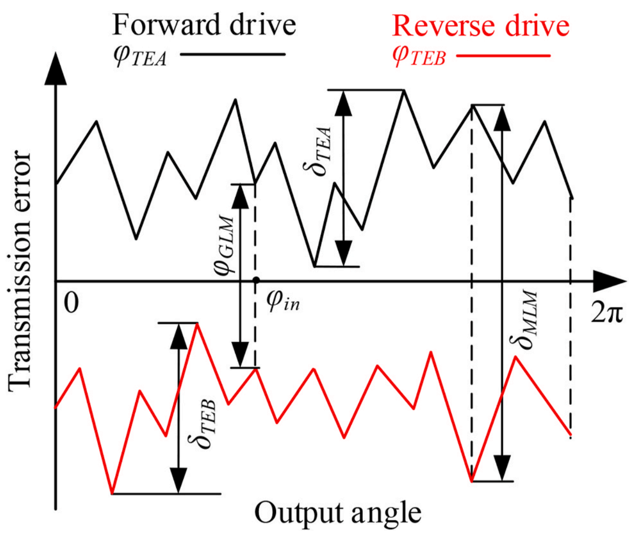

Fig. 3. Schematic diagram of the BDDTE of the entire reducer.

indicators. The forward drive TE is φ_{TEA}, and its variation range is δ_{TEA}. The reverse drive TE is φ_{TEB}, and its variation range is δ_{TEB}. Since the cycloid pin gear transmission and pin gear are connected in series, as shown in Eq. (6), the forward drive TE φ_{TEA} of the entire reducer is the sum of the TE φ_{TECA} of the cycloid gear and the TE φ_{TETA} of the output mechanism. As shown in Eq. (7), the reverse drive TE φ_{TEB} of the reducer is the sum of the TE φ_{TECB} of the cycloid gear and the TE φ_{TETB} of the output mechanism. As shown in Eq. (8), the difference between φ_{TEA} and φ_{TEB} is defined as the global lost motion (GLM) [54] of the reducer. The calculation result of φ_{GLM} contains the LM information of any output angle position of the reducer, and let the maximum value of φ_{GLM} be δ_{MLM}.

$$\varphi_{TEA} = \varphi_{TECA} + \varphi_{TETA} \tag{6}$$

$$\varphi_{TEB} = \varphi_{TECB} + \varphi_{TETB} \tag{7}$$

$$\varphi_{GLM} = \varphi_{TEA} - \varphi_{TEB} \tag{8}$$

The positioning accuracy requirements [1,55] for precision cycloid reducers are usually the TE of no more than 60 arcsecs and the LM of no more than 90 arcsecs. Numerous studies [9,36,43,52] have demonstrated that the cycloid reducer's TE has nonlinear time-varying characteristics. The conventional TE analysis only considers the forward or reverse drive TE and does not study the difference between them. At the same time, some studies have conducted LM tests [42–44] on the reducer and found that the LM at different output angular positions is not equal. The maximum and minimum value deviations are large. Thus, the precision reducer's positioning accuracy must consider the BDDTE and the GLM. The positioning accuracy requirements of the precision cycloid reducer are shown in Eq. (9).

$$\begin{cases} 0 \le \delta_{TEA} \le 60 \text{ arcsecs} \\ 0 \le \delta_{TEB} \le 60 \text{ arcsecs} \\ 0 \le \delta_{MLM} \le 90 \text{ arcsecs} \end{cases} \tag{9}$$

2.3. New assumption for equivalence modeling of MSEs

The main components that affect the reducer's positioning accuracy are the pin gear, the cycloid gear, and the output flange. The components are affected by MSEs [37], including various manufacturing and assembly errors. As shown in Fig. 4, the pins distribution radius error δ_{Rp}, the pin radius error δ_{rp}, and the pin gear tooth dividing error λ_p may affect the actual position of the pins in the pin gear. The crankshaft eccentricity error δ_e, the cycloid gear center hole eccentricity error (δ_{cr}, ζ_{cr}), the cycloid gear tooth dividing error λ_c, and the cycloid gear tooth profile error δ_{cp} may affect the actual position of cycloid gear tooth profile. The cycloid gear pin holes distribution radius error δ_{Rw}, the pin hole radius error δ_{rw}, and the pin hole dividing error λ_w may affect the assembly relationship between the cycloid gear and the output flange. The output flange pins distribution radius error δ_{Rs}, the pin sleeve radius error δ_{rs}, and the pin dividing error λ_w may affect the actual position of the output flange.

The modeling of the center positions of the pins is commonly simplified by assuming the radial distribution of each pin is exactly the same [36,42,47]. However, the center position of the pin is affected by the coupling effect of multiple factors, such as the pin gear tooth groove profile error, the pin radius error, and the pin assembly error. The pin gear tooth groove is an internal tooth, which is commonly manufactured by single-tooth grinding [56] of the forming wheel. The outer diameter of the designed pin gear is 70 mm, and the diameter of the grinding wheel rod for grinding the internal teeth is limited, and the radial stiffness of the grinding wheel rod is relatively low, so the radial runout values of different pin gear tooth groove may be different, i.e., the values of δ_{Rpi} are not the same.

In addition, there are several different equivalent modeling assumptions for the pin gear tooth dividing error λ_p and the cycloid gear

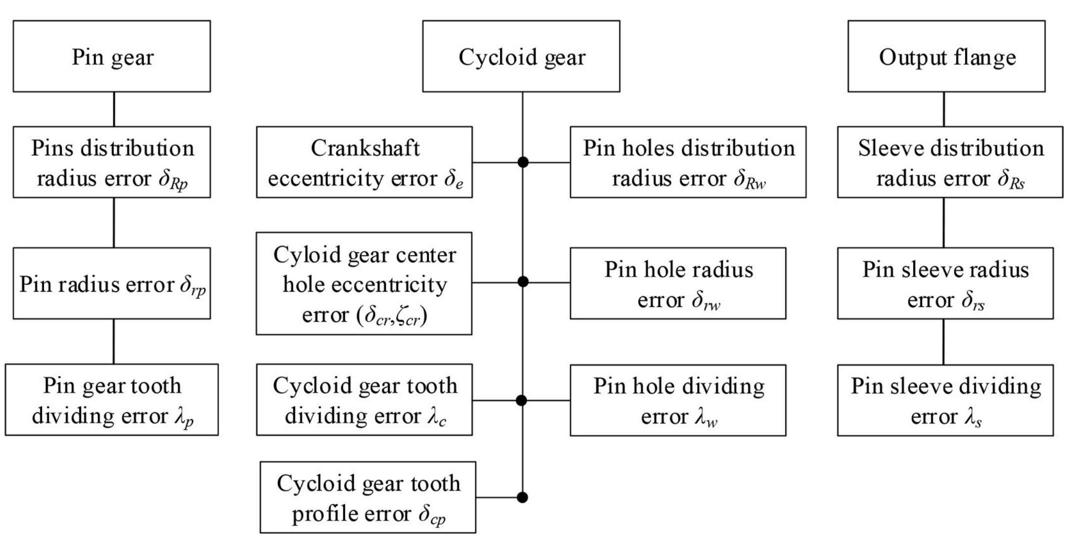

Fig. 4. Multi-source errors of main components.

tooth dividing error λ_c. The influence of this error parameter λ_p on the positioning accuracy of the reducer is not considered in the literature [32]. In the literature [36], it is assumed that the numerical distribution of λ_p satisfies the same sinusoidal distribution as the accumulated tooth pitch error of general gears. In the literature [57], it was found that the cycloid gear tooth indexing error λ_c approximately satisfies the sinusoidal distribution by measurement. In contrast, in the literature [58], it was found that the pin gear tooth dividing error λ_p does not satisfy the sinusoidal distribution by measurement. The main reason for the deviation of the above assumptions from the measured conclusions of the component is the influence of the actual manufacturing process on the geometric characteristics of the MSEs.

In this study, the indexing devices of the selected processing equipment to manufacture components adopt closed-loop control, so the angular indexing of the pin gear tooth will not produce cumulative errors [59]. However, due to the possible uneven roughing allowance and the influence of workpiece clamping position errors, the force on each tooth during single-tooth profile grinding may be nonlinear, so the values of λ_p and λ_c may be closer to normal distribution. Since different MSE equivalence modeling assumptions directly impact the positioning accuracy prediction results of the reducer, this study will introduce a new MSE measurement scheme in Section 4 and validate the proposed MSE equivalence modeling assumptions experimentally.

2.4. Geometry characteristics modeling of MSEs

Fig. 5 shows the geometric characteristics of the MSEs of the single-stage cycloid drive. The point O_{pi} in Fig. 5(a) is the ideal position of the i-th pin, and the point O_{pti} is the position of the i-th pin affected by the pin distribution radius error δ_{Rpi} and the pin tooth indexing error λ_{pi}. The position (x_{Oei}, y_{Oei}) of the pin center in the fixed pin gear coordinate system $X_pO_pY_p$ is expressed as Eq. (10). The pin radius error δ_{rpi} affects the size of the pin. O_c is the ideal position of the cycloid gear rotation center, O_{ct} is the position of the cycloid gear rotation center affected by the crankshaft eccentricity error δ_e, and the coordinates of O_{ct} under $X_pO_pY_p$ are calculated by Eq. (11). As shown in Fig. 5(b), the tooth profile of the j-th cycloid gear is affected by the cycloid gear tooth indexing error λ_{cj} and the cycloid gear center hole eccentricity error (δ_{cr}, ζ_{cr}). In $X_pO_pY_p$, the tooth profile coordinates (x_{cmpj}, y_{cmpj}) of j-th cycloid gear tooth are expressed as Eq. (12). δ_{cr} is the center hole eccentricity error, ζ_{cr} is the phase angle parameter of the eccentricity, and (x_{cmj}, y_{cmj}) is the actual manufactured cycloid gear tooth profile coordinates in $X_cO_cY_c$.

As shown in Fig. 5(c), the manufactured tooth profile is the curve between the upper deviation equidistant curve and the lower deviation equidistant curve of the designed tooth profile, and the profile error of the left and right flank is usually not symmetrical. Assume that the actual manufacturing tooth profile differs from the design tooth profile by δ_{cpN} at the N-th meshing point in the normal direction ($N = 1, 2, \ldots, N_c$). The tooth profile errors are related to the grinding wheel dressing error [60], and it is difficult to carry out forward modeling of the tooth profile error in the design stage. However, when the cycloid gear tooth is processed by single-tooth profile grinding, it can be considered that the profile errors of each tooth are exactly the same. Therefore, the

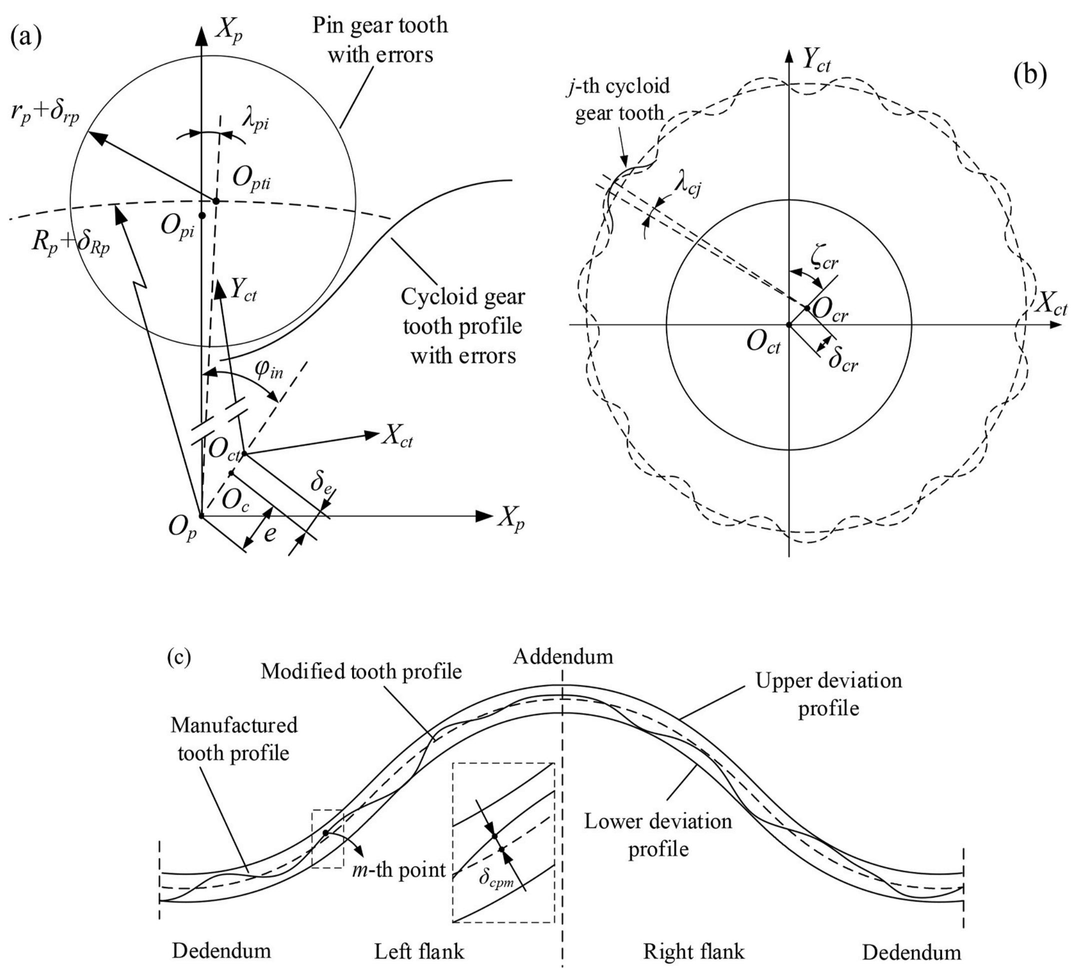

Fig. 5. Geometric characteristics of (a) MSEs of cycloid drive (b) cycloid gear tooth indexing error and center hole eccentricity error (c) cycloid gear tooth profile error.

equivalent modeling of the tooth profile error can be realized by measuring the manufactured cycloid gear. Through the Fourier fitting of the measurement results, the equivalent modeling of δ_{cpN} is realized, which is expressed as Eq. (13). N is the harmonic order, a and b_n are the amplitudes of harmonic orders, and φ_c is the tooth profile design parameter.

$$\begin{cases} x_{Oei} = \left(R_p + \delta_{Rpi}\right)\sin\left(\dfrac{2\pi(1-i)}{Z_p} + \lambda_{pi}\right) \\ y_{Oei} = \left(R_p + \delta_{Rpi}\right)\cos\left(\dfrac{2\pi(1-i)}{Z_p} + \lambda_{pi}\right) \end{cases} \tag{10}$$

$$\begin{cases} x_{Oct} = (e+\delta_e)\sin\varphi_{in} \\ y_{Oct} = (e+\delta_e)\cos\varphi_{in} \end{cases} \tag{11}$$

$$\begin{cases} x_{cmpj} = \left(x_{cmj}+\delta_{cr}\sin\xi_{cr}\right)\cos\left(\varphi_{out}+\lambda_{cj}\right) - \left(y_{cmj}+\delta_{cr}\cos\xi_{cr}\right)\sin\left(\varphi_{out}+\lambda_{cj}\right) + x_{Oct} \\ y_{cmpj} = \left(x_{cmj}+\delta_{cr}\sin\xi_{cr}\right)\sin\left(\varphi_{out}+\lambda_{cj}\right) + \left(y_{cmj}+\delta_{cr}\cos\xi_{cr}\right)\cos\left(\varphi_{out}+\lambda_{cj}\right) + y_{Oct} \end{cases} \tag{12}$$

$$\begin{cases} x_{spk} = (R_w+\delta_{Rw})\sin\left(\dfrac{2\pi(1-k)}{Z_s}+\lambda_{wk}\right)\cos\varphi_{out} - (R_w+\delta_{Rw})\cos\left(\dfrac{2\pi(1-k)}{Z_s}+\lambda_{wk}\right)\sin\varphi_{out} \\ y_{spk} = (R_w+\delta_{Rw})\sin\left(\dfrac{2\pi(1-k)}{Z_s}+\lambda_{wk}\right)\sin\varphi_{out} + (R_w+\delta_{Rw})\cos\left(\dfrac{2\pi(1-k)}{Z_s}+\lambda_{wk}\right)\cos\varphi_{out} \end{cases} \tag{17}$$

$$\delta_{cpN} = A_0 + \sum_{n=1}^{\infty}\left[a_n\cos(n\omega\varphi_c) + b_n\sin(n\omega\varphi_c)\right] \tag{13}$$

Define the relationship between the normal direction ε_i of the meshing point and the design parameter φ_{cj} of the j-th cycloid gear profile as $\varepsilon(\varphi_{cj})$, then under the influence of the indexing error λ_{cj} and the tooth profile error δ_{cpN}, the j-th cycloid tooth profile (x_{cmj}, y_{cmj}) in $X_cO_cY_c$ is expressed as Eq. (14). The value range of parameter φ_{cj} is $\varphi_{cj} \in$ [-π/Z_c+2 (1-j)π/Z_c, π/Z_c+2 (1-j)π/Z_c]. (x_{cdj}, y_{cdj}) is the tooth profile with modification design.

$$\begin{cases} x_{cmj} = x_{cdj} - \delta_{cpN}\sin\varepsilon\left(\varphi_{cj}\right) \\ y_{cmj} = y_{cdj} + \delta_{cpN}\cos\varepsilon\left(\varphi_{cj}\right) \end{cases} \tag{14}$$

Fig. 6 shows geometric characteristics of the multi-source errors of the output mechanism when the input angle is φ_{in}. The subscript k is used to distinguish different values of the same error parameter, k = 1,2, …, Z_s. The coordinates (x_{wck}, y_{wck}) of the k-th pin hole center O_{wtk} in $X_cO_cY_c$ can be expressed as Eq. (15). The coordinates (x_{wpk}, y_{wpk}) of O_{wtk} in $X_pO_pY_p$ can be expressed as Eq. (16). Under the influence of the pin hole radius error δ_{rwk}, the actual radius of the k-th pin hole is r_w+δ_{rwk}.

$$\begin{cases} x_{wck} = (R_w+\delta_{Rwk})\sin\left(\dfrac{2\pi(1-k)}{Z_s}+\lambda_{wk}\right) \\ y_{wck} = (R_w+\delta_{Rwk})\cos\left(\dfrac{2\pi(1-k)}{Z_s}+\lambda_{wk}\right) \end{cases} \tag{15}$$

$$\begin{cases} x_{wpk} = x_{wck}\cos\varphi_{out} - y_{wck}\sin\varphi_{out} + x_{Oct} \\ y_{wpk} = x_{wck}\cos\varphi_{out} + y_{wck}\sin\varphi_{out} + y_{Oct} \end{cases} \tag{16}$$

Similarly, the pin distribution radius error δ_{Rsk} and the pin indexing error λ_{wk} affect the actual position of the pin shaft on the output flange. When the input angle is φ_{in}, the coordinates (x_{spk}, y_{spk}) of the k-th pin center O_{stk} in $X_pO_pY_p$ is expressed as Eq. (17). Under the influence of the pin sleeve radius error δ_{rsk}, the actual radius of the k-th pin sleeve is r_s+δ_{rsk}. φ_{out} in the above equations is the ideal output angle, and $\varphi_{out} = \varphi_{in}/Z_c$.

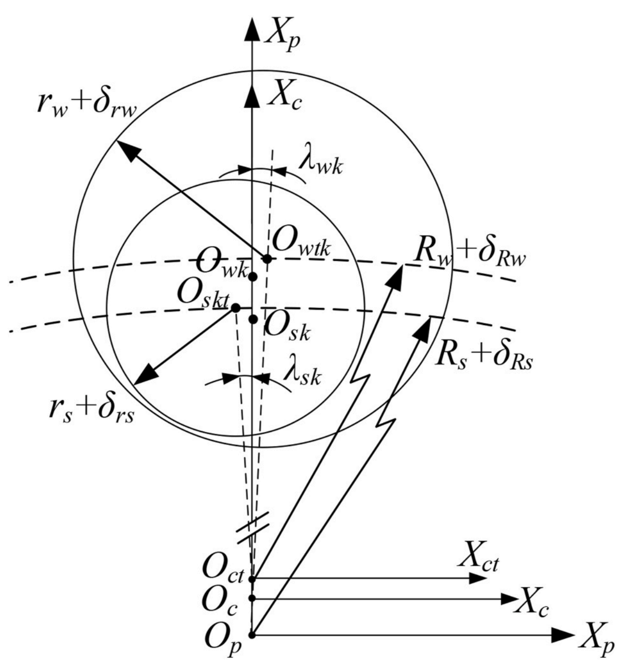

Fig. 6. Geometric characteristics of MSEs of the output mechanism.

3. BDDTE model for positioning accuracy prediction of the entire reducer

3.1. BDDTE model for the single-stage cycloid drive

Due to the influence of MSEs, the cycloid gear and pin gear are no longer in an ideal conjugate relationship, and the BDDTE model of the cycloid drive is established. Take the forward drive as an example, and

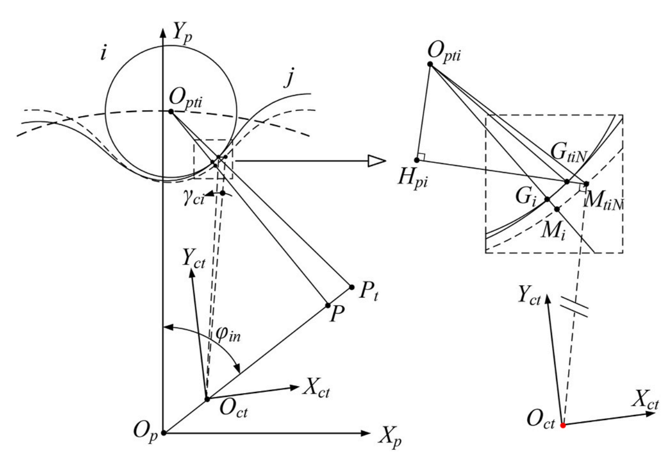

Fig. 7. Coordinate systems defined for cycloid drive TE analysis.

the pin gear coordinate system $X_pO_pY_p$ is set as the fixed coordinate system. In Fig. 7, the input angle is φ_{in}, and P is the node of the ideal cycloid drive. Under ideal conditions, the point M_i on the cycloid gear will mesh with the point G_i on the i-th pin. In reality, there is an initial clearance of unknown size between each cycloid gear tooth and the pin, and the cycloid gear needs to rotate around the rotational center O_{ct} before it can contact the pin. The conversion from the cycloid gear tooth profile coordinates to the pin gear coordinate system is calculated by Eq. (18)–(20), assuming that the point M_{tiN} on the j-th cycloid gear tooth profile is rotated by γ_{ci} and contacts the i-th pin on the point G_{tiN}. The subscript N is the number of discrete points on the tooth profile of the cycloid gear, $N = 1, 2, \ldots, N_c$, and N_c is the number of discrete points on the left flank of the cycloid gear tooth. Since the rotation angle γ_{ci} is small, $G_{tiN}M_{tiN} \perp O_{ct}M_{tiN}$ can be approximated according to the geometric relationship. Therefore, the vertical line of $G_{tiN}M_{tiN}$ can be made by O_{pti}, and the vertical foot is H_p. Then γ_{ci} is calculated by Eq. (21), and the parameters in Eq. (21) are expressed as Eq. (22). The rotation angle of the cycloid gear is min (γ_{ci}), $i = 1, 2, \ldots, Z_p$. According to the definition min (γ_{ci}) is the deviation between the actual output angle and the theoretical output angle, that is, the forward drive TE φ_{TECA} of the cycloid pin gear pair. Similarly, based on this model, the reverse drive TE φ_{TECB} of the cycloid pin gear pair can also be analyzed. Finally, φ_{TECA} and φ_{TECB} are expressed as Eq. (23).

$$r_c(\varphi_{in}) = \left[x_{cmpj}, y_{cmpj}, 0, 1\right]^{\mathrm{T}} \tag{18}$$

$$R_m = \begin{bmatrix} \cos\gamma_m & -\sin\gamma_m & 0 & (1-\cos\gamma_m)(e+\delta_e)\sin\varphi_{in} + (e+\delta_e)\cos\varphi_{in}\sin\gamma_m \\ \sin\gamma_m & \cos\gamma_m & 0 & (1-\cos\gamma_m)(e+\delta_e)\cos\varphi_{in} - (e+\delta_e)\sin\varphi_{in}\sin\gamma_m \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{19}$$

$$r_{pcm}(\varphi_{in}) = \left[x_{cmptj}, y_{cmptj}, 0, 1\right]^T = R_m r_c(\varphi_{in}) \tag{20}$$

$$\gamma_{ci} = \min\left(\overline{G_{ti}M_{tiN}} \,/\, \overline{O_{ct}M_{tiN}}\right), (N = 1, 2, \cdots, N_c) \tag{21}$$

$$\begin{cases} \overline{G_{tiN}M_{tiN}} = \overline{H_pM_{tiN}} - \overline{H_pG_{tiN}} \\ \overline{H_pM_{tiN}} = \sqrt{\overline{O_{pti}M_{tiN}}^2 - \overline{O_{pti}H_p}^2} \\ \overline{H_pG_{tiN}} = \sqrt{(r_p+\delta_{rp})^2 - \overline{O_{pti}H_p}^2} \end{cases} \tag{22}$$

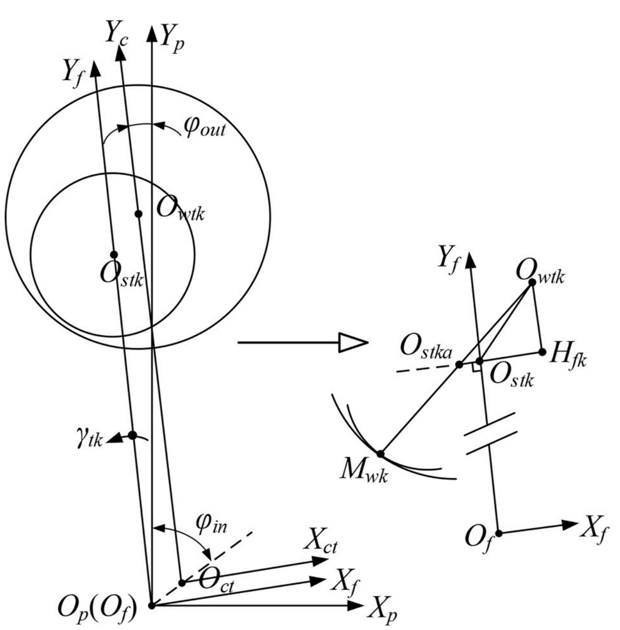

Fig. 8. Coordinate systems defined for output mechanism TE analysis.

$$\begin{cases} \varphi_{TECA} = \min(\gamma_{ci}), \left(i = 1, 2, \cdots, Z_p/2\right) \\ \varphi_{TECB} = -\min(\gamma_{ci}), \left(i = Z_p/2+1, Z_p/2+2, \cdots, Z_p\right) \end{cases} \tag{23}$$

3.2. BDDTE model for the output mechanism

Similarly, a BDDTE model of the output mechanism can also be established. Take the forward drive as an example, and the pin gear coordinate system $X_pO_pY_p$ is set as the fixed coordinate system. In Fig. 8, the input angle is φ_{in}, and $X_fO_fY_f$ is the rotation of the output flange. O_{stk} is the actual center point of the k-th pin shaft with error, O_{wtk} is the actual center point of the k-th pin hole with error, and there will also be an initial clearance of unknown size between each pin and the pin hole. Assuming that after the output flange is rotated by an angle γ_{ti}, the k-th pin shaft is in contact with the M_{wk} point on the k-th pin hole, and the center point moves from O_{stk} to O_{stka}. It is the same as the rotary contact mechanism of the cycloid pin gear pair. Since the rotation angle γ_{ti} is small, it can be approximated as $O_{stka}O_{stk} \perp O_{stk}O_f$ according to the geometric relationship. Therefore, the vertical line of $O_{stka}O_{stk}$ can be made from O_{wtk}, and the vertical foot is H_{fk}. Then γ_{tk} is calculated by Eq. (24), and the parameters in Eq. (24) are expressed as Eq. (25). The rotation angle of the output flange is min (γ_{tk}), $k = 1, 2, \ldots, Z_s$. According to the definition min (γ_{tk}) is the forward drive TE φ_{TETA} of the output mechanism. Similarly, based on this model, the reverse drive TE φ_{TETB} of the output mechanism can also be analyzed. Finally, φ_{TETA} and φ_{TETB} are expressed as Eq. (26).

$$\gamma_{tk} = \overline{O_{stka}O_{stk}} \big/ \overline{O_{stk}O_f} \tag{24}$$

$$\begin{cases} \overline{O_{stka}O_{stk}} = \overline{O_{stka}H_{fk}} - \overline{O_{stk}H_{fk}} \\ \overline{O_{stka}H_{fk}} = \sqrt{(r_w+\delta_{rwk} - (r_s+\delta_{rsk}))^2 - \overline{O_{wtk}H_{fk}}^2} \\ \overline{O_{stk}H_{fk}} = \sqrt{\overline{O_{stk}O_{wtk}}^2 - \overline{O_{wtk}H_{fk}}^2} \end{cases} \tag{25}$$

$$\begin{cases} \varphi_{TETA} = \min(\gamma_{tk}), (k = 1, 2, \cdots, Z_w/2) \\ \varphi_{TETB} = -\min(\gamma_{tk}), (k = Z_w/2+1, Z_w/2+2, \cdots, Z_w) \end{cases} \tag{26}$$

3.3. BDDTE model for the entire reducer

Based on the BDDTE model of the cycloid drive and the output mechanism, the positioning accuracy of the entire reducer can be analyzed by the following steps.

(1) Set the initial step number of the input angle as $m = 1$, the input angle step size as $\Delta\varphi_{in}$, then the input shaft rotation angle as $\varphi_{in} = m\Delta\varphi_{in}$. The forward and reverse drive TE is analyzed by the BDDTE models in Section 3.1 and Section 3.2, and the BDDTE of the entire reducer is calculated by Eq. (6) and Eq. (7).
(2) Increase the number of steps of the input shaft rotation angle m, update the coordinate position of the components, and continuously analyze the BDDTE. If $\varphi_{TEA} < \varphi_{TEB}$, the cycloid gear, and the pin gear have assembly interference, the analysis must be terminated.
(3) Increase the input angle until the output angle $\varphi_{out} \geq 2\pi$. The GLM of the entire reducer can be calculated by Eq. (8).

The positioning accuracy of the cycloid reducer is affected by the manufacturing and assembly error of both the cycloid pin gear pair and the output mechanism. Therefore, to realize the effective prediction of the positioning accuracy of the reducer before assembly, it is necessary to further study the tolerance allocation and the MSEs measurement to provide a theoretical basis for improving the reducer positioning accuracy.

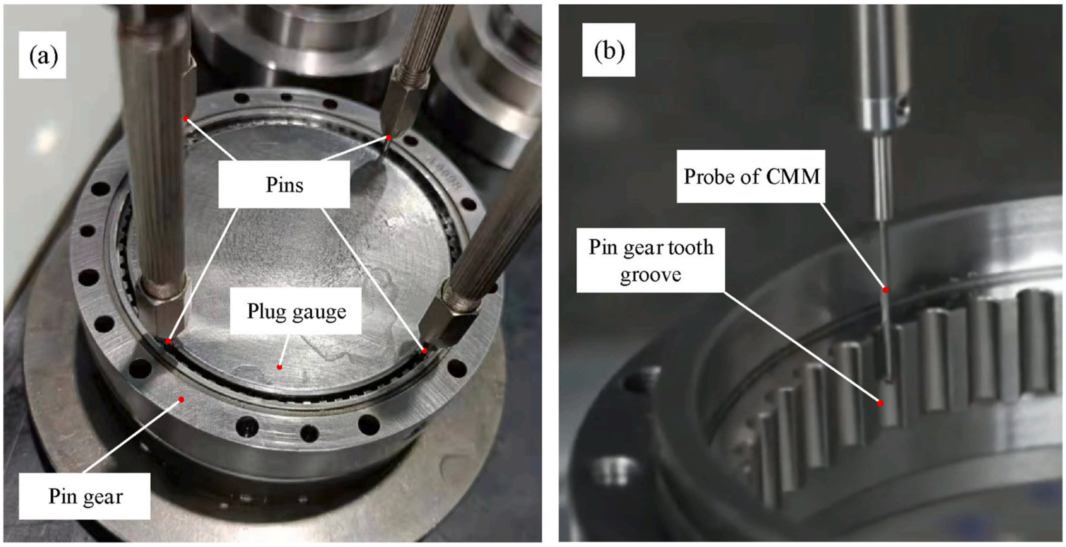

Fig. 9. Schematic diagram of the conventional pin gear MSEs measurement of (*a*) first solution (b) second solution.

4. Positioning accuracy prediction and improvement design of the entire reducer

4.1. MSEs measurement of main components

To verify the rationality of the proposed MSE equivalence modeling assumptions, it is necessary to fabricate parts according to the tolerance design parameters in Table 1 and measure them for MSEs. At present, the conventional pin gear error measurement scheme is mainly divided into two kinds. The first solution is shown in Fig. 9(a), where a standard plug gauge of known diameter is placed in the pin gear. Select four pins with the same outside diameter and four equally spaced pin gear tooth groove mounting positions, and place different sizes of plug gauges until four pins can be installed into the grooves. The disadvantage of the first solution is that the standard of the pins assembly depends on the working experience of each operator. There may be deviations in the standards of different operators, which may affect the measurement results. At the same time, the pin gear distribution radius error δ_{Rp} measured by this solution is a constant value and cannot describe the radial position deviation of different pins. The second solution is shown in Fig. 9(b), where a CMM is used to measure the pin gear tooth groove profile, and the radial and tangential position deviations of each tooth groove are calculated by matching with the design results. The advantage of the second solution over the first one is that the description of the radial and tangential errors of the pins can be achieved. However, the second solution is an indirect measurement, and its disadvantage is that the measurement is only the error of the tooth groove and does not take into account the actual installation condition of different sizes of pins in each tooth groove.

To address the above-mentioned problems, an equivalent measurement method for the radial and tangential error of the pin center position of the pin gear is presented. As shown in Fig. 10, the pin gear workpiece is clamped on the closed-loop controlled precision rotary table, O_d is the center of rotation of the precision rotary table, O_m is the reference center of the pin gear, and the red line is the horizontal line. Considering the clamping error of the pin gear workpiece, there is a positional deviation between O_d and O_m, so it is necessary to first establish a reference for pin gear error measurement. To this end, as shown in Fig. 10(a), four pins with the same diameter are installed in the groove of the pin gear at 90-degree intervals, and the pins are numbered ABCD. The center point of each pin is used as a reference point, in which the center point of pin A, for example, is O_{pA}. The main steps required for

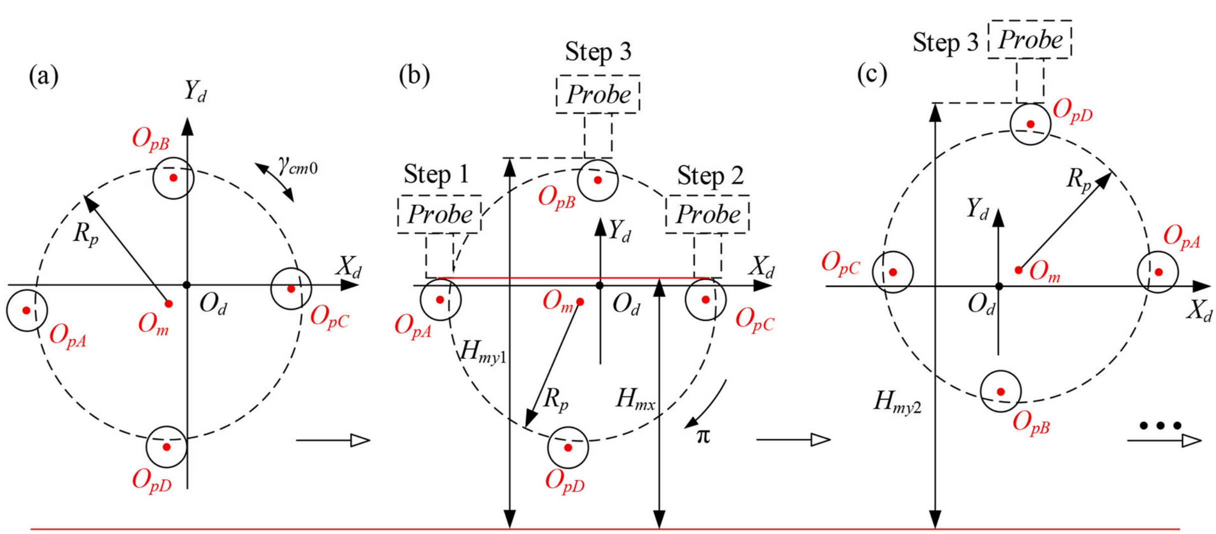

Fig. 10. (a) Schematic diagram of pin gear clamping error (b) Steps 1 to 3 of measurement reference selection (c) Step 4 of measurement reference selection.

the measurement are as follows.

(1) Step 1: Measure the height of the upper surface of the pin located at 9 o'clock.
(2) Step 2: Measure the height of the upper surface of the pin at 3 o'clock.
(3) Step 3: Measure the height of the upper surface of the pin at 12 o'clock.

The specific measurement process is as follows, first of all, as shown in Fig. 10(b), rotate the precision rotary table several times to repeat Step 1 and Step 2 until the height measurement results of pin A and pin C are the same value H_{mx}. Assuming that the upper surface of pin A and pin C is in the horizontal line, and the angle of the rotary table at this time is set to 0°. Use Step 3 to measure the height H_{my1} of pin B. Next, as shown in Fig. 10(c), rotate the rotary table 180° counterclockwise and use measure the height H_{my2} of pin D by Step 3. After that, rotate the rotary table 90° counterclockwise and measure the height H_{my3} of pin A by Step 3. Finally, rotate the rotary table 90° counterclockwise again and measure the height H_{my4} of pin C by Step 3. At this point, the exact height of the four pins at 12 o'clock has been measured. Based on this, the reference center O_m for pin gear error measurement can be established by the above steps, where the horizontal deviation between O_m and O_d is $(H_{my1} + H_{my2})/2\text{-}R_p$, and the vertical deviation between O_m and O_d is $(H_{my3} + H_{my4})/2\text{-}R_p$.

On the basis of establishing the measurement reference center O_m, the actual assembly error of each pin of the pin gear in the tooth groove can be measured. The measurement principle of the radial runout error of the pin circle is shown in Fig. 11(a). The pin number in the 12 o'clock direction is No. 1, and the pin numbers increase in the counterclockwise direction. First, measure the height HRp1 of the No. 1 pin, and then rotate $2\pi/Z_p$ to measure the height H_{Rp2} of the No. 2 pin. Rotate $2\pi/Z_p$ in turn and measure each pin height H_{Rpi}. Through the conversion with the basic coordinates of O_m, the radial runout of each pin tooth, that is, the value of the distribution radius error δ_{Rpi} of the actual pin center, is obtained.

The measurement principle of the tangential angle indexing error of the pin is shown in Fig. 11(b), and the pin numbering method is exactly the same as that of the δ_{Rpi} value measurement. Measure the height $H_{\lambda pi}$ of the i-th pin in the direction of 9 o'clock from the initial position, and then rotate $2\pi/Z_p$ in turn to measure the height of each pin. Through the conversion with the basic Om coordinates, the tangential indexing error of each pin, that is, the value of the pin indexing error λ_{pi} is obtained.

The measurement process of the pin gear MSEs is shown in Fig. 12 (a). The pin installed in the tooth groove is measured by a Mitutoyo MDH-25MB external micrometer. The error of the pin radius is shown in Table A1. The workpiece is clamped on a precision closed-loop rotary table equipped with a 21-bit grating encoder, and the error measurement uses a three-coordinate measuring instrument of the Chienwei CWT-544 A V-CNC model. The measurement process of the cycloid gear MSEs is shown in Fig. 12(b), and the error measurement uses a three-coordinate measuring instrument of the Werth VideoCheck-S model. The tooth profile error of each cycloid gear tooth is approximately the same. The red curve in Fig. 12(c) is the measured tooth profile error, and the black curve is the measured result fitted by Eq. (13). The fitting result is shown in Eq. (27) and Table 2. The measurement results show that the accuracy of the modified cycloid gear profile meets the tolerance design requirements.

$$\delta_{cpkN} = 0.014 + \sum_{n=1}^{6} [a_n \cos(48.3n\varphi) + b_n \sin(48.3n\varphi)] \tag{27}$$

The manufacturing and assembly errors of the main components are all measured by precision measuring equipment. The results are shown in Appendix A1, A2, and A3, including pin distribution radius error δ_{Rp}, pin radius error δ_{rp}, pin tooth indexing error λ_p, and cycloid gear tooth indexing error λ_c, etc. The measurement result of crankshaft eccentricity error is $\delta_e = -2.0$ μm, cycloid gear hole eccentricity error is $\delta_{cr} = 1.2$ μm, cycloid gear hole eccentricity error phase angle is $\zeta_{ro} = 318.6$ °, cycloid gear pin hole distribution radius error is $\delta_{Rw} = 0.5$ μm, pin hole radius error is $\delta_{rw} = 4.4$ μm, and output flange pin distribution radius error is $\delta_{Rs} = 1.8$ μm. The measurement results of the distribution radius error δ_{Rpi} of different numbers of pins in Table A1 are not the same, and the deviation between the maximum value and the minimum value exceeds 4 μm, which does not belong to the category of measurement error. Therefore, the measurement results verify the same deceleration proposed in this study. The assumption is that the pin distribution radius error δ_{Rpi} in the machine prototype is not equal in value. Similarly, neither the indexing error λ_p of the pin tooth of the pin gear nor the indexing error λ_c of the cycloid gear tooth in Table A2 satisfies the sinusoidal distribution law, which verifies the rationality of ignoring the cumulative pitch error proposed in this research based on the analysis of the manufacturing process.

4.2. Positioning accuracy simulation based on Monte Carlo method

Due to the uncertainty of most of the manufacturing and assembly errors in the design stage, the Monte Carlo method [61] is commonly used. Monte Carlo method is a numerical calculation method based on random sampling, which starts from a probabilistic statistical model and simulates a large number of random samples by computer to obtain the statistical characteristics of the quantity to be solved and use it as an approximate solution to the problem. For this study, the Monte Carlo method can be used to equivalently model the error values of the reducer components under the condition of the initial tolerance distribution scheme, and through a large number of virtual prototypes, the analysis of the positioning accuracy realizes the prediction of the variation range of the positioning accuracy of the actual manufacturing

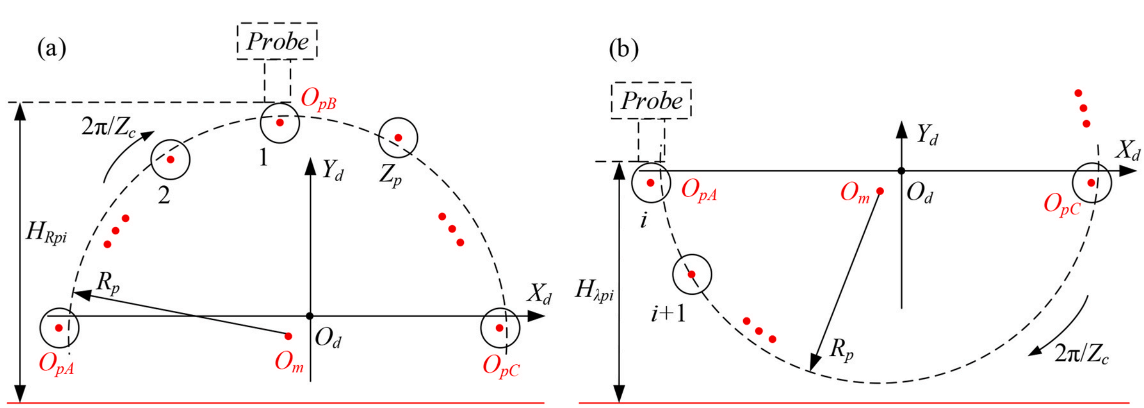

Fig. 11. Schematic diagram of measurement principle of (a) pin distribution radius error δ_{Rpi} (b) pin indexing error λ_{pi}.

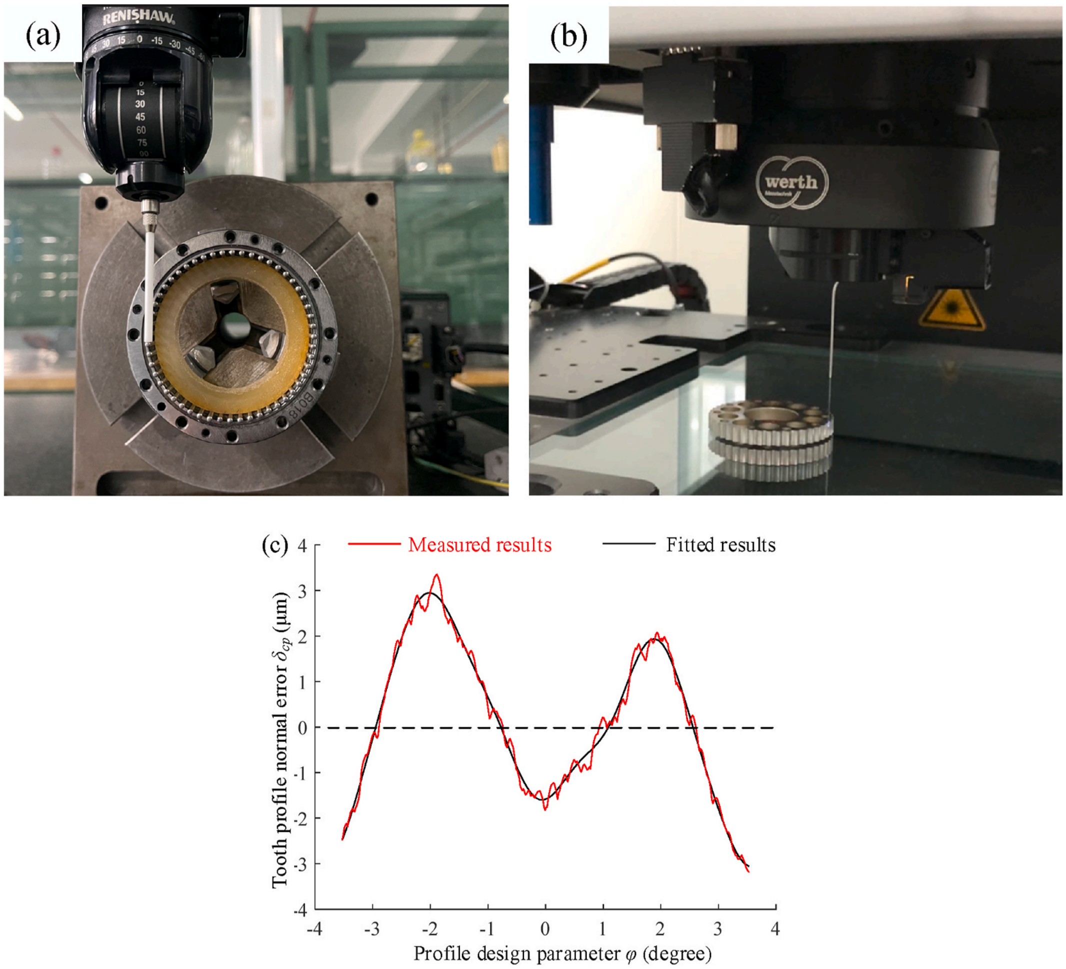

Fig. 12. MSEs measurement (a) of pin gear (b) of cycloid gear (c) results of cycloid tooth profile error.

Table 2
Fitting parameters.

n	1	2	3	4	5	6
a_n	0.42	−2.23	−0.33	0.05	−0.07	−0.10
b_n	−0.66	−0.29	−0.12	0.07	0.08	0.06

prototype. Assuming that the number of virtual prototypes is N_q, the cycloid gear tooth profile error is equivalently modeled by Eq. (27). The eccentricity error δ_e of the crankshaft and the eccentricity error δ_{cr} of the center hole of the cycloid gear belongs to the eccentricity error. When the number of N_q is large, the values of δ_e and δ_{cr} follow the Rayleigh distribution within the tolerance range, the values of ζ_{cr} follow the mean distribution, and the rest of the MSEs values follow the standard overall distribution within the tolerance range. Based on Latin hypercube sampling [62], the MSE values are equivalently modeled, and the positioning accuracies of N_q virtual prototypes are simulated using the BDDTE model. Fig. 13 shows the simulation results of 30,000 virtual prototypes, and the probability of assembly interference among the 30, 000 virtual prototypes is 1.8 %. Since the probability of assembly interference is low, the tolerance allocation scheme in Table 1 is reasonable.

As shown in Fig. 13(a), the 99.7 % confidence interval of the forward drive δ_{TEA} of the reducer is [29.7,73.9] arcsecs, and the probability of the single index δ_{TEA} satisfying the precision transmission performance of Eq. (9) is 93.0 %. Let the forward drive TE of the kinematic pair be δ_{TECA}. Fig. 13(b) shows the ratio of the forward drive TE of the single-stage cycloid drive to the forward drive TE of the entire reducer $\delta_{TECA}/\delta_{TEA}$. This parameter provides a description of the influence of the single-stage cycloid drive on the forward drive TE of the entire reducer. The 99.7 % confidence interval of $\delta_{TECA}/\delta_{TEA}$ was [51.9 %, 109.9 %], and the mean value of $\delta_{TECA}/\delta_{TEA}$ was 77.4 %. The results show that there is a large deviation between the positioning accuracy of the cycloid drive and the positioning accuracy of the entire reducer, and the influence of the TE of the output mechanism on the performance of the entire reducer cannot be ignored. Some of the $\delta_{TECA}/\delta_{TEA}$ values greater than 100 % indicate that there is a complex nonlinear coupling between the TE of the output mechanism and the TE of the single-stage cycloid drive.

As shown in Fig. 13(c), the 99.7 % confidence interval of δ_{TEB} for the reverse transmission of the reducer is [32.9,82.4] arcsecs, and the probability that the single index δ_{TEB} satisfies the precision transmission performance of Eq. (9) is 83.0 %. The reason for the large deviation between the reverse drive TE and the forward drive TE prediction is that the cycloidal tooth profile error is asymmetrical. Let the reverse drive TE of the cycloidal kinematic pair be δ_{TECB}, the 99.7 % confidence interval of $\delta_{TECB}/\delta_{TEB}$ in Fig. 13(d) is [53.7 %, 108.2 %], and the average value of $\delta_{TECB}/\delta_{TEB}$ is 77.5 %. This result is in good agreement with the analysis results of the forward drive TE, which also verifies that the influence of the output mechanism on the TE of the entire reducer is strong.

As shown in Fig. 13(e), the 99.7 % confidence interval of the maximum GLM δ_{MLM} of the reducer is [113.2, 243.9] arcsecs, and the probability of the single index δ_{MLM} satisfying the requirements of Eq. (9) is 0 %. Obviously, the GLM performance of the reducer is more difficult to meet the precision reducer requirements than the TE. Let the maximum geometric hysteresis of the cycloid kinematic pair be δ_{MLMC}, the 99.7 % confidence interval of $\delta_{MLMC}/\delta_{MLM}$ in Fig. 13(f) is [33.7 %,

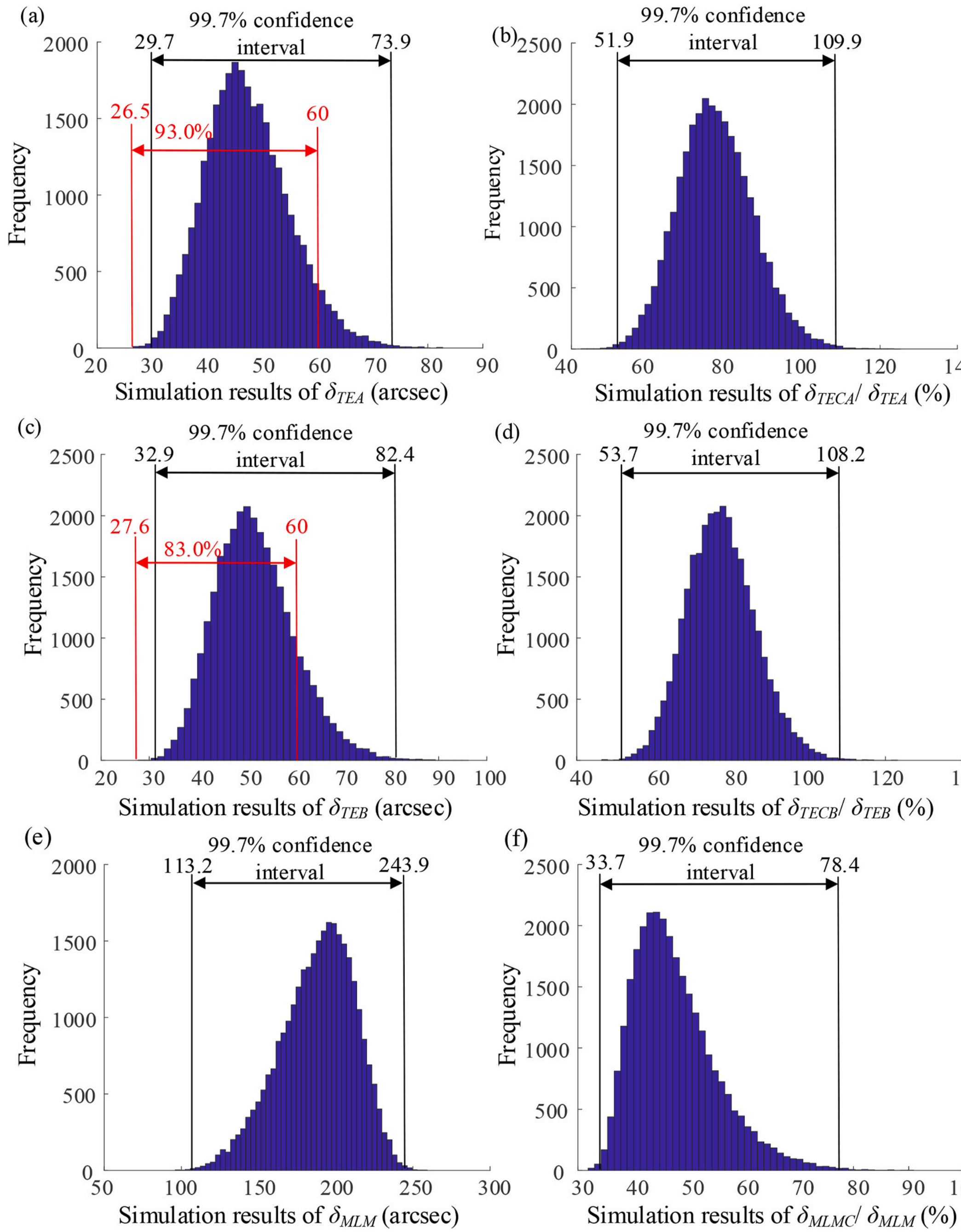

Fig. 13. Monte Carlo simulation results of: (a) forward cycloid drive TE δ_{TECA} (b) percentage of $\delta_{TECA}/\delta_{TEA}$ (c) reverse cycloid drive TE δ_{TECB} (d) percentage of $\delta_{TECB}/\delta_{TEB}$ (e) the maximum GLM of cycloid drive δ_{MLMC} (f) the percentage of $\delta_{MLMC}/\delta_{MLM}$.

78.4 %], and the average value of $\delta_{TECB}/\delta_{TEB}$ is 47.6 %. The results show that the output mechanism has a stronger effect on the maximum GLM of the entire reducer than the single-stage cycloid drive.

4.3. Positioning accuracy improvement design solution

From the Monte Carlo simulation results, it can be found that due to the uncertainty of the manufacturing error value, there is an obvious gap between the positioning accuracy of the virtual prototype and the performance requirements, especially since the GLM of the virtual prototype is far from the requirement of the precision reducer for 90 arcsecs. The most effective solution to reduce the GLM is to optimize the tooth-side clearance between each pin tooth. According to the studies [24,47,48,63], when the tooth-side clearance between each pin tooth is reduced by the same value at the same time, the influence on the GLM is linear. However, since the deviation between the maximum and minimum values of δ_{MLM} in the virtual prototypes exceeds 130 arcsecs while reducing the GLM of some of the virtual prototypes, the virtual prototypes with the small GLM will produce assembly interference. Therefore, the optimization of the tooth-side clearance alone cannot make all virtual prototypes meet the positioning accuracy requirements without reducing the assembly success rate. In the reported works, the improvement of the performance of the reducer is usually studied only for the single-stage cycloid drive, and there is little research on the performance improvement of the output mechanism. Therefore, it is necessary to consider the transmission characteristics of the cycloid drive and the output mechanism to study the performance improvement of the entire reducer.

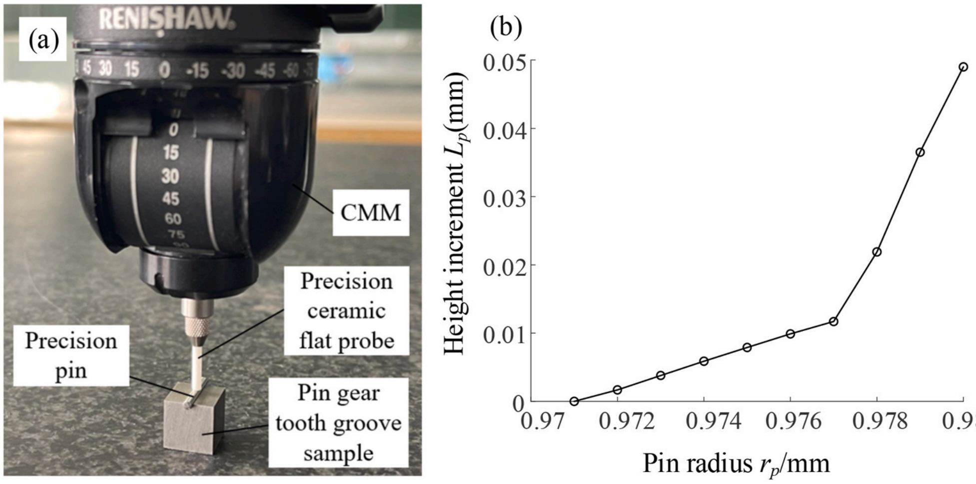

Fig. 14. (a) pin gear tooth groove sample (b) pin assembly error measurement results.

The positioning accuracy of the reducer is influenced by the assembly clearance of both the cycloid drive and the output mechanism. The assembly clearance of the cycloid gear and pin gear can be actively adjusted by the measurement and option of the pins, while the assembly clearance of the output mechanism parts can be actively adjusted by the measurement and option of the pin sleeve. Since the finished pin gear groove profile error cannot be changed and the change of pin radius will cause the change of radial position of the pin center, it is necessary to study the mapping relationship between pin radius and pin position change through measurement test and correct the actual pin center position after pin matching. Based on the tolerance parameters in Table 1, the grinding wheel of the pin gear internal teeth grinding is dressed, and a sample of the pin gear tooth groove is produced. The measurement process and measured results of the assembly error of the pin in the tooth groove are shown in Fig. 14(a) and (b). Ten pins are screened out at intervals of 1 μm within the range of radius [0.971, 0.980] mm by using the Mitutoyo MDH-25MB outer diameter micrometer with a resolution of 0.1 μm, and the pins are installed in the tooth groove in turn. The height of the upper surface of the pins installed in the tooth groove is measured with a Chienwei CWT-544 CMM with 0.1 μm resolution. Let the height of the upper surface of the pin with the smallest radius be 0 mm. Take the pin radius r_p as the X axis and the upper surface height increment ΔL_p as the Y axis. As shown in Fig. 14(b), the value of ΔL_p changes abruptly when the radius of the pin is greater than 0.977 mm. The measurement results show that the pin radius error has a certain influence on the center position of the installed pin in the pin gear tooth groove. The diameter of the pin installed in the tooth groove should be less than 1.954 mm. Otherwise, the distribution radius of the installed pins will be greatly reduced, which will easily cause assembly interference. When the diameter of the pin is less than 1.954 mm, the change in the distribution radius of the pin center caused by the pin radius error δ_{rp} can be approximated as $-1.0\delta_{rp}$ through the linear fitting.

The positioning accuracy of the entire reducer is affected by the assembly clearance of both the single-stage cycloid drive and the output mechanism, so this study proposes an active control scheme of the assembly clearance based on the MSE measurement of the key components and the matching. In the first step, for the cycloid drive tooth side clearance, the tolerance distribution range of the pin radius error is gradually shifted to positive tolerance until the BDDTE analysis and inspection of the single-stage cycloid drive produces assembly interference. Let the final tolerance offset be U_{rp}. In the second step, according to the contact clearance of the output mechanism, the size of the pin sleeve installed on each pin shaft is optimized.

According to the Monte Carlo analysis results based on the BDDTE model, the performance of the TE of the reducer is easier to meet the requirements of the precision reducer than the GLM, so the optimized objective function and constraints are shown in Eq. (28), where δ_{rsk} is the radius error of the k-th pin sleeves ($k = 1,2, \ldots,10$). Considering the actual measurement and matching situation, the discrete particle swarm algorithm (DPSO) [64] is used to solve the problem, and the value of the discrete degree is 0.1 μm. The DPSO method is an optimization algorithm based on group intelligence, which is an extension of the particle swarm optimization algorithm on discrete problems. The basic idea of the method is to represent the feasible solution of the problem as a group of discrete particles, each of which has a certain velocity and position, and to approach the optimal solution of the problem step by step by continuously updating the velocity and position of the particles. During the updating process, each particle adjusts its speed and position according to its own experience and the experience of the swarm to achieve better search results. Based on this scheme, theoretically, the positioning accuracy of the reducer can be effectively improved, but the actual performance improvement effect needs to be verified by comparative tests based on the selection and matching of physical prototype components.

$$\begin{cases} \min f\left(U_{rp}, \delta_{rsk}\right) = \delta_{MLM} \\ 0\ \mu\text{m} \le U_{rp} \le 10\ \mu\text{m} \\ -10\ \mu\text{m} \le \delta_{rsk} \le 10\ \mu\text{m} \\ 0 \text{ arcsec} \le \delta_{TEA} \le 60 \text{ arcsecs} \\ 0 \text{ arcsec} \le \delta_{TEB} \le 60 \text{ arcsecs} \end{cases} \tag{28}$$

5. Positioning accuracy tests of the prototype

5.1. Positioning accuracy prediction of the prototype

The MSEs measurement results of each component are brought into the BDDTE model to analyze the positioning accuracy of the reducer. The analysis results are shown in Fig. 15(a). The forward drive TE is 57.6 arcsecs, the reverse drive TE is 62.3 arcsecs, and the maximum GLM is 144.5 arcsecs. The analysis results show that the positioning accuracy of the reducer prototype needs to be improved. Based on the scheme introduced in Section 4.4, the tolerance range of the pin radius is optimized. The tolerance range of the optimized error δ_{rp} is [0,1.5] μm. The optimized matching dimensions of each pin sleeve are shown in Appendix A4. The predicted performance of the improved prototype is shown in Fig. 15(b). The forward drive TE of the optimized reducer is 51.2 arcsecs, the reverse drive TE is 52.1 arcsecs, and the maximum GLM is 76.8 arcsecs. All positioning accuracy indexes of the optimized

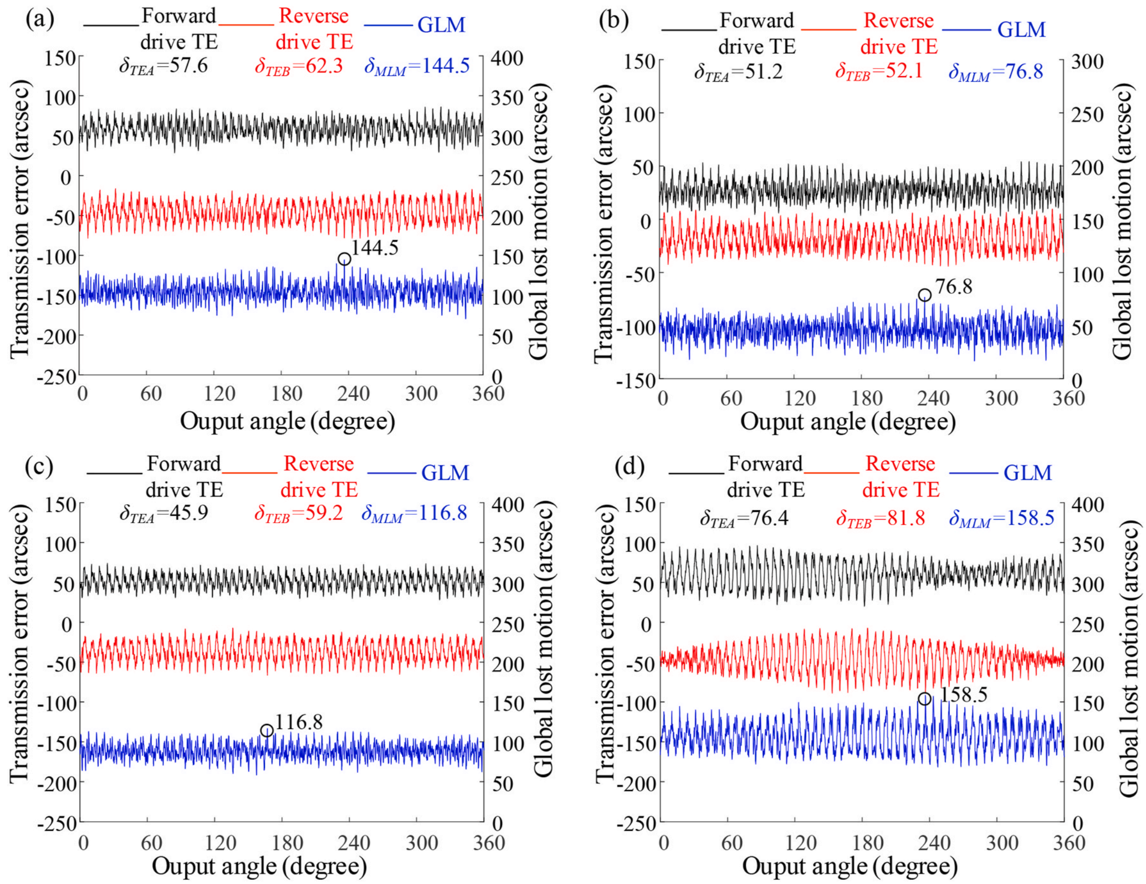

Fig. 15. Prediction results of positioning accuracy of the (a) prototype based on the assumptions made in this study (b) improved prototype (c) Case 1 based on different assumptions (d) Case 2 based on different assumptions.

reducer meet the performance requirements of the precision reducer. Therefore, the proposed positioning accuracy improvement solution can theoretically achieve significant improvement in the performance of the reducer, especially the GLM performance.

To study the influence of different equivalent modeling assumptions on the prediction results of the positioning accuracy of the reducer, the discussion is carried out based on specific calculation cases. The prediction results of Case 1 are shown in Fig. 15(c), where it is assumed that the pin distribution radius error δ_{Rp} is a constant value, and the rest of the MSE parameters are exactly the same as those calculated in Fig. 15(a). The value of δ_{Rp} is obtained by the conventional measurement method in Fig. 9(a), and the measurement result is $\delta_{Rp} = 0.006$ mm. The prediction results of Case 1 show that the forward drive TE is 45.9 arcsecs, the reverse drive TE is 59.2 arcsecs, and the maximum GLM is 116.8 arcsecs. The analysis results of Case 2 are shown in Fig. 15(d), where it is assumed that the pin gear tooth indexing error λ_c satisfies the sinusoidal distribution in the range of [−15,15] arcsecs, and the rest of the MSE parameters are exactly the same as those calculated in Fig. 15

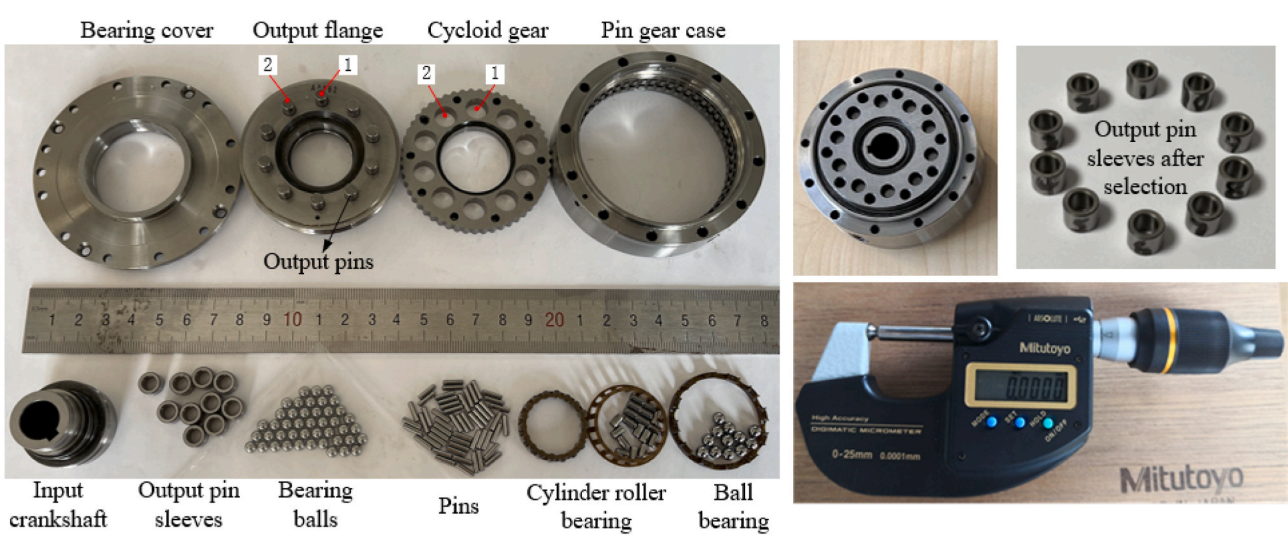

Fig. 16. Reducer prototype and pins after measurement and sorting.

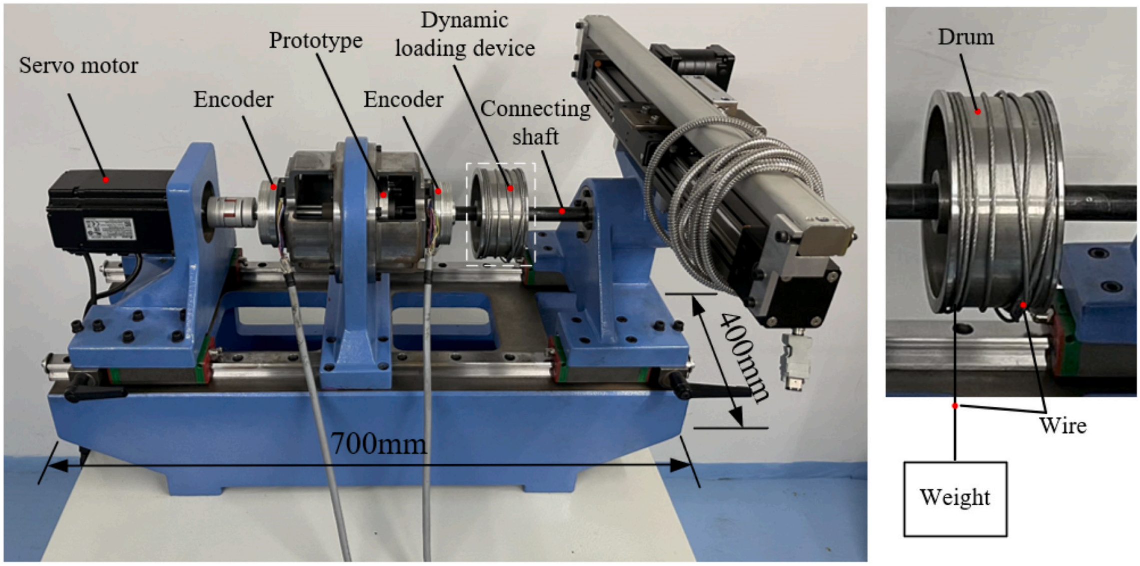

Fig. 17. Precision reducer test rig.

(a). The prediction results of Case 2 show that the forward drive TE of the reducer is 76.4 arcsecs, the reverse drive TE is 81.8 arcsecs, and the maximum GLM is 158.5 arcsecs. From the analysis results in Fig. 15(a), (c), and Fig. 15(d), it can be found that the predicted positioning accuracy of the reducer based on different assumptions of multi-source error equivalence modeling has a large deviation. Therefore, it is necessary to carry out test experiments for verification.

5.2. Tests of the prototype

Fig. 16 contains all the components used to assemble the prototype, the assembled prototype, the micrometer for measurement, and the pin sleeves selected after the improvement design. The outer diameter of the reducer housing is 80 mm, and the rated load of the reducer is T_c = 18Nm. Then, a precision reducer comprehensive performance test rig is designed, which can perform the BDDTE test of the prototype.

The test rig is shown in Fig. 17. The input side and output side of the tested reducer are fitted with high-precision grating encoders. The connecting shaft is connected to the output flange of the reducer and is supported by bearings. During the TE test, a load of −3 % or +3 % rated torque can be added to the output flange of the reducer by the dynamic loading device, which consists of the drum, the wire, and the weight. The drum is permanently attached to the connecting shaft, the wire is installed in the outer raceway of the drum, and the weight is installed at the end of the wire. When the output of the reducer rotates at a constant speed, the weight is subjected to gravity which produces a certain moment on the drum. It is known that 3 % of the rated load of the reducer is 0.54 Nm and the radius of the drum is 250 mm. Let the acceleration of gravity be 9.8 N/kg, the weight to be added is about 440 g. The dynamic torque obtained by this approach is more stable than the torque obtained by the classic magnetic powder brake.

The BDDTE test steps of the reducer prototype are as follows.

(1) Adding a counterclockwise load of 0.54 Nm to the output flange of the reducer through the weight and setting the value of the encoder at the output side to zero.
(2) The servo motor rotates at a constant speed clockwise, the input angle is recorded as θ_{in}, and the output angle is recorded as θ_{out}. When θ_{out} is greater than 360°, the servo motor stops after one more rotation. The test result of the forward drive TE in the range of output angle [0, 2π] can be calculated by Eq. (29).
(3) Adding a clockwise load of 0.54 Nm to the output flange, the servo motor rotates in reverse. The servo motor stops when the output angle returns to zero. The reverse drive TE test result can be calculated by Eq. (30). The test results of the prototype's GLM can be calculated by Eq. (31).

$$TEA_{\text{test}} = \theta_{out} - \frac{\theta_{in}}{Z_c} \tag{29}$$

$$TEB_{\text{test}} = \theta_{out} - \frac{\theta_{in}}{Z_c} \tag{30}$$

$$GLM_{\text{test}} = TEA_{\text{test}} - TEB_{\text{test}} \tag{31}$$

During the test, the data sampling frequency is 50 Hz, and the input speed of the servo motor is set to 15 rpm to reduce the influence of high-speed impacts. The precision rotary grating encoder used on the test rig is 20 bits, its resolution is 1.2 arcsec, and the absolute error of measurement is ±3 arcsecs. In order to further verify the rationality of the GLM test results, it is necessary to perform multiple hysteresis curve tests at different output angle positions. Based on the BDDTE test method, the positioning accuracy of the reducer prototype is tested before and after the component matching.

5.3. Tests results and discussion

The BDDTE test results of the initially assembled reducer prototype are shown in Fig. 18(a). The black line is the test result of the forward drive TE, and the maximum variation range of the forward drive TE is TEA_{test} = 61.8 arcsecs. The red line is the test result of the reverse drive TE, and the maximum variation range of the reverse drive TE is TEB_{test} = 65.9 arcsecs, and the TE performance of the prototype is close to the performance requirements of the precision cycloid reducer. The blue line is the result of the GLM test, and the maximum value of the GLM is MLM_{test} = 153.3 arcsecs. The test results in Fig. 18(a) are compared with the different theoretical prediction results in Fig. 16, and the deviation statistics of the prediction results are shown in Table 3.

As shown in Table 3, the influence of different equivalent modeling assumptions on the positioning progress prediction results is very obvious. The prediction results in Fig. 15(a) obtained by the BDDTE model based on the new assumption proposed in this study are closest to the prototype test results compared with the conventional assumptions. The deviation of the forward drive TE of the reducer from the predicted results of the proposed model is about −6.8 %, the deviation of the

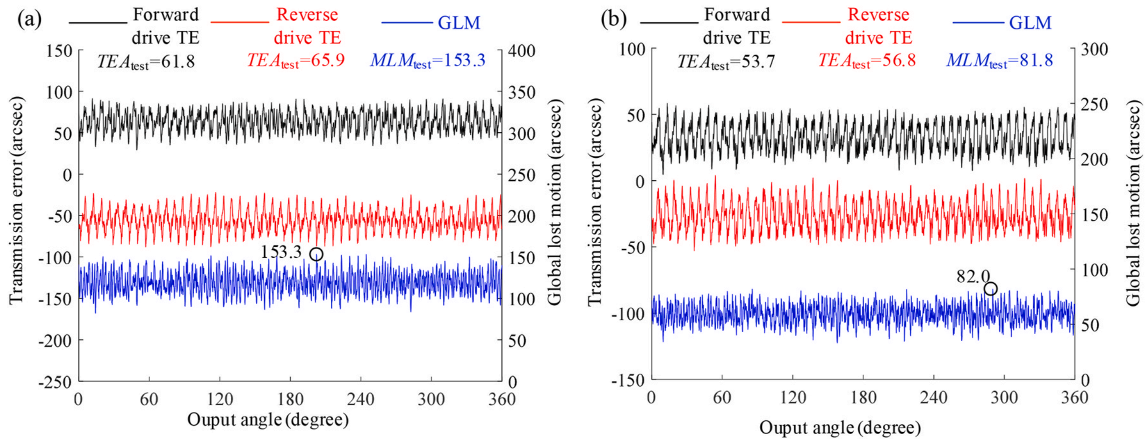

Fig. 18. Test results (a) before improvement (b) after improvement.

Table 3
Deviations of predicted and tested results of positioning accuracy.

Parameters	Deviation of δ_{TEA} from TEA_{test}	Deviation of δ_{TEA} from TEB_{test}	Deviation of δ_{MLM} from MLM_{test}
Prediction results of the proposed model	−6.8 %	−5.5 %	−5.7 %
Prediction results of Case 1	−25.7 %	−10.2 %	23.8 %
Prediction results of Case 2	23.6 %	24.1 %	3.4 %

reverse drive TE is about −5.5 %, and the deviation of the maximum GLM is about −5.7 %. Since the absolute accuracy of the encoder is ±3.0 arcsec, the deviation is acceptable.

The deviations between the test results and the prediction results of Case 1 in Fig. 15(c) are large, with the deviations for both the forward drive TE and the maximum GLM exceeding 20 %. The deviation of the maximum GLM in Case 2 and the test result is small, but the deviation of the predicted forward drive TE and reverse drive TE exceeds 20 %. The comparison results of Case 2 show that unreasonable equivalence modeling assumptions may produce a very close match prediction due to the chance of MSE values. Therefore, for the prediction of the positioning accuracy of the reducer, the forward drive TE, reverse drive TE, and GLM must be taken into account.

The BDDTE test results of the optimized reducer prototype after components matching are shown in Fig. 18(b). The forward drive TE is TEA_{test} = 53.7 arcsecs, the reverse drive TE is TEB_{test} = 56.8 arcsecs, and the maximum GLM is MLM_{test} = 81.8 arcsecs. The optimized prototype fully meets the performance requirements of the precision reducer, and the test results of the optimized reducer are in high agreement with the theoretical prediction results in Fig. 16(b).

In summary, the prototype test results before and after the improvement verify the accuracy of the proposed BDDTE model for the positioning accuracy prediction of the entire cycloid reducer. The test results of the prototype after the improvement verify the effectiveness of the proposed positioning accuracy improvement design solution.

6. Conclusions

In this study, a bi-directional drive transmission error (BDDTE) model for the analysis of the positioning accuracy of the entire cycloid reducer is established considering the influence of actual MSEs, based on which the forward drive transmission error (TE), reverse drive TE, and global lost motion (GLM) of the reducer can be predicted at the same time. The main findings and contributions of this study are summarized as follows.

(1) The proposed MSE equivalent modeling assumption of pin gear is closer to the actual situation than the other assumptions. The BDDTE model based on this assumption has a higher accuracy in predicting the positioning accuracy of the entire reducer. Therefore, it has a better guiding effect on the tolerance allocation and also helps reduce the interference probability of the prototype assembly.
(2) The results of Monte Carlo analysis show that the positioning accuracy of the single-stage cycloid drive mechanism is not completely dominant in the positioning accuracy of the entire reducer, and the impact of the output mechanism on the positioning accuracy of the entire reducer cannot be ignored. The results of the prediction of positioning accuracy show that for the evaluation of the positioning accuracy of precision cycloid reducers, the performance of forward drive TE, reverse drive TE, and GLM must be considered simultaneously. If only unidirectional TE or only GLM performance is considered, a performance misjudgment of the reducer may occur.
(3) The proposed improvement design solution for the positioning accuracy of the cycloid reducer is effective. Therefore, this study provides certain theoretical and practical guidance for improving the product performance of precision cycloid reducers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1

Table A1
Components measurement results of δ_{rpi} and δ_{Rpi}

i	1	2	3	4	5	6	7	8	9	10	11
δ_{rpi} (μm)	−0.7	−0.8	−1	−0.8	−1	−1.1	−0.6	−0.4	−0.9	−1.2	−0.3
δ_{Rpi} (μm)	9.4	7.2	8.8	5.6	7.6	7.9	7.6	6.6	6.9	9.1	8.3
i	12	13	14	15	16	17	18	19	20	21	22
δ_{rpi} (μm)	−1	−0.9	−0.9	−0.8	−1	−0.8	−0.7	−0.9	−0.5	−0.7	−0.9
δ_{Rpi} (μm)	8.0	7.4	7.8	6.9	6.7	6.9	5.5	8.7	9.6	7.7	8.4
i	23	24	25	26	27	28	29	30	31	32	33
δ_{rpi} (μm)	−0.9	−1.0	−0.8	−0.9	−1.0	−0.9	−0.7	−0.9	−0.9	−0.8	−0.7
δ_{Rpi} (μm)	8.3	9.4	9.8	8.5	7.4	7.2	8.0	8.3	6.5	8.0	8.1
i	34	35	36	37	38	39	40	41	42	43	44
δ_{rpi} (μm)	−1.1	−1.1	−0.6	−0.9	−0.9	−0.4	−0.8	−0.9	−0.8	−0.7	−0.6
δ_{Rpi} (μm)	8.8	8.0	7.4	7.9	6.2	9.5	6.3	8.6	7.2	8.7	9.9
i	45	46	47	48	49	50	51	52			
δ_{rpi} (μm)	−0.7	−0.6	−0.6	−0.7	−0.8	−0.7	−0.6	−0.6			
δ_{Rpi} (μm)	7.8	8.4	9.0	7.7	7.8	8.3	9.2	6.9			

Table A2
Components measurement results of λ_{pi} and λ_{cj}

i	1	2	3	4	5	6	7	8	9	10	11
λ_{pi} (arcsec)	−3.5	−0.2	−4.2	6.1	−1.8	0.3	0.2	−2.6	−2.9	1.8	−1.5
λ_{cj} (arcsec)	4.0	5.6	8.6	−2.2	1.5	0.9	−1.0	3.4	−0.5	−5.8	−4.4
i	12	13	14	15	16	17	18	19	20	21	22
λ_{pi} (arcsec)	3.5	−2.8	4.0	2.7	−8.0	1.5	−0.5	1.4	4.9	5.0	−1.5
λ_{cj} (arcsec)	0.8	−6.4	2.3	−3.6	2.2	−1.0	0.6	−4.0	6.5	2.0	1.6
i	23	24	25	26	27	28	29	30	31	32	33
λ_{pi} (arcsec)	0.3	−4.2	−2.3	1.5	−0.3	−1.3	3.2	−4.8	5.4	−0.6	−12.0
λ_{cj} (arcsec)	1.1	1.9	0.7	1.2	2.7	−2.8	6.4	−2.3	1.6	−5.6	−3.9
i	34	35	36	37	38	39	40	41	42	43	44
λ_{pi} (arcsec)	−0.5	−1.9	−2.3	−7.8	9.6	3.4	5	−5.1	0.1	4.0	4.4
λ_{cj} (arcsec)	0.1	−10.6	−5.4	2.7	1.3	0.4	8.3	1.7	1.2	8.9	−0.7
i	45	46	47	48	49	50	51	52			
λ_{pi} (arcsec)	−1.2	−2.7	−6.2	−0.5	0.6	−0.8	2.8	−4.0			
λ_{cj} (arcsec)	−1.3	2.4	−2.1	−0.6	2.1	8.3	10.6				

Table A3
Components measurement results of δ_{sk}、λ_{sk} and λ_{wk}

k	1	2	3	4	5	6	7	8	9	10
δ_{sk} (μm)	1.6	0.5	1.1	0.7	1	0.9	1.2	1.3	1	0.8
λ_{wk} (arcsec)	5.5	4.9	−11.3	0.4	−1.7	−0.8	6.5	4.3	−1.1	−1.7
λ_{sk} (arcsec)	6.5	7.9	1.6	−3.8	−0.3	−1.5	1.4	8	3.2	−8.7

Table A4
Values of δ_{sk} after optimization

k	1	2	3	4	5	6	7	8	9	10
δ_{sk} (μm)	3.1	2.9	2.2	2.1	2.7	3.0	3.1	2.7	2.5	2.7

References

[1] Bilancia P, Monari L, Raffaeli R, Peruzzini M, Pellicciari M. Accurate transmission performance evaluation of servo-mechanisms for robots. Robot Comput Integrated Manuf 2022;78:102400.
[2] Huijun Y, Xiangkai W, Yao Z, Yue S, Yong Z, Lintao Y, et al. A comprehensive cycloid PIN-wheel precision reducer test platform integrated with a new dynamic measurement method of lost motion. Metrol Meas Syst 2022;29:207–29.
[3] Jiang N, Wang S, Xie X, Yuan X, Yang A, Zhang J. A vectorial modification methodology based on an efficient and accurate cycloid tooth profile model. Precis Eng 2022;73:435–56.
[4] Litvin FL, Feng PH. Computerized design and generation of cycloidal gearings. Mech Mach Theor 1996;31:891–911.
[5] Hong-Sen Yan, Ta-Shi, Lai. Geometry design of an elementary planetary gear train with cylindrical tooth-profiles. Mech Mach Theor 2002;37:757–67.
[6] Li X, He W, Li L, Schmidt LC. A new cycloid drive with high-load capacity and high efficiency. J Mech Des 2004;126:683.
[7] Sheu KB, Chien CW, Chiou ST, Lai TS. Kinetostatic analysis of a roller drive. Mech Mach Theor 2004;39:819–37.
[8] Lai T-S. Geometric design of roller drives with cylindrical meshing elements. Mech Mach Theor 2005;40:55–67.
[9] Chen B, Zhong H, Liu J, Li C, Fang T. Generation and investigation of a new cycloid drive with double contact. Mech Mach Theor 2012;49:270–83.
[10] Hsieh CF. Traditional versus improved designs for cycloidal speed reducers with a small tooth difference: the effect on dynamics. Mech Mach Theor 2015;86:15–35.

[11] Demenego A, Vecchiato D, Litvin FL, Nervegna N, Mancó S. Design and simulation of meshing of a cycloidal pump. Mech Mach Theor 2002;37:311–32.
[12] Teterukov DO. Method for backlash and load distribution among pins determination of trochoid transmission with account machining tolerances. Proc World Congr Mech Mach Sci 2004:843–7.
[13] Meng Y, Wu C, Ling L. Mathematical modeling of the transmission performance of 2K–H pin cycloid planetary mechanism. Mech Mach Theor 2007;42:776–90.
[14] Gorla C, Davoli P, Rosa F, Longoni C, Chiozzi F, Samarani A. Theoretical and experimental analysis of a cycloidal speed reducer. J Mech Des 2008:130.
[15] Zhang Y, Li L, Ji S. Influence of cycloid-pin gear design parameters on bearing capacity and optimized design. J Braz Soc Mech Sci Eng 2022;44:123.
[16] Li L, Guan T, Wang Z, Liu Y, Wei Y, Wan C. The computer aided design of cycloid drive. Chin J Mech Eng 1990;4:221–31.
[17] Hong-Liu Y, Jin-Hua Y, Xin H, Ping S. Study on teeth profile modification of cycloid reducer based on non-Hertz elastic contact analysis. Mech Res Commun 2013;48: 87–92.
[18] Wei B, Wang JX, Zhou GW, Yang RS, Zhou HJ, He T. Mixed lubrication analysis of modified cycloidal gear used in the RV reducer. Proc Inst Mech Eng Part J J Eng Tribol 2016;230:121–34.
[19] Sun X, Han L, Wang J. Tooth modification and loaded tooth contact analysis of China Bearing Reducer. Proc Inst Mech Eng Part C J Mech Eng Sci 2019;233: 6240–61.
[20] Yang DCH, Blanche JG. Design and application guidelines for cycloid drives with machining tolerances. Mech Mach Theor 1990;25:487–501.
[21] Hidaka T, Wang HY, Ishida T, Matsumoto K, Hashimoto M. Rotational transmission error of K-H-V planetary gears with cycloid gear : 1st report, analytical method of the rotational transmission error. Trans Jpn Soc Mech Eng Ser C 1994;60:645–53.
[22] Ishida T, Wang H, Hidaka T, Hashimoto M. Rotational Transmission Error of K-H-V-Type planetary gears with cycloid gears: 2nd report, effects of manufacturing and assembly errors on rotational transmission error. Trans Jpn Soc Mech Eng Ser C 1994;60:3510–7.
[23] Wang H, Ishida T, Hidaka T, Hashimoto M. Rotational transmission error of K-H-V-type planetary gears with cycloid gear : 3rd report, mutual effects of errors of the elements on the rotational transmission error. Trans Jpn Soc Mech Eng Ser C 1994; 60:3518–25.
[24] Han L, Guo F. Global sensitivity analysis of transmission accuracy for RV-type cycloid-pin drive. J Mech Sci Technol 2016;30:1225–31.
[25] Hsieh CF. The effect on dynamics of using a new transmission design for eccentric speed reducers. Mech Mach Theor 2014;80:1–16.
[26] Xu L, Yang Y. Dynamic modeling and contact analysis of a cycloid-pin gear mechanism with a turning arm cylindrical roller bearing. Mech Mach Theor 2016; 104:327–49.
[27] Hsieh C-F, Fuentes-Aznar A. Performance prediction method of cycloidal speed reducers. J Braz Soc Mech Sci Eng 2019:41.
[28] Xuan L, Xie C, Guan T, Lei L, Jiang H. Research on dynamic modeling and simulation verification of a new type of FT pin-cycloid transmission. Proc Inst Mech Eng, Part C: J Mech Eng Sci 2019;233:6276–88.
[29] Tung L-C, Chan YJ. A time-variant dynamic model for compound epicyclic–cycloidal reducers. Mech Mach Theor 2023;179:105095.
[30] Sun Y, Zhao X, Jiang F, Zhao L, Liu D, Yu G. Backlash analysis of RV reducer based on error factor sensitivity and monte-carlo simulation. Int J Hybrid Info Tech 2014; 7:283–92.
[31] Lin W-S, Shih Y-P, Lee J-J. Design of a two-stage cycloidal gear reducer with tooth modifications. Mech Mach Theor 2014;79:184–97.
[32] Li X, Li C, Wang Y, Chen B, Lim T. Analysis of a cycloid speed reducer considering tooth profile modification and clearance-fit output mechanism. J Mech Des 2017; 139:033303.
[33] Ren ZY, Mao SM, Guo WC, Guo Z. Tooth modification and dynamic performance of the cycloidal drive. Mech Syst Signal Process 2017;85:857–66.
[34] Xu H, Shi Z, Yu B, Wang H. Optimal measurement speed and its determination method in the transmission precision evaluation of precision reducers. Appl Sci 2019;9:2146.
[35] Li T, Tian M, Xu H, Deng X, Su J. Meshing contact analysis of cycloidal-pin gear in RV reducer considering the influence of manufacturing error. J Braz Soc Mech Sci 2020;42:1–14.
[36] Yang Y, Zhou G, Chang L, Chen G. A modelling approach for kinematic equivalent mechanism and rotational transmission error of RV reducer. Mech Mach Theor 2021;163:104384.
[37] Xu LX, Zhong JL, Li Y, Chang L. Design and dynamic transmission error analysis of a new type of cycloidal-pin reducer with a rotatable output-pin mechanism. Mech Mach Theor 2023;181:105218.
[38] Tariq H, Galym Z, Amrin A, Spitas C. Assessment of contact forces and stresses, torque ripple and efficiency of a cycloidal gear drive and its involute kinematical equivalent. Mech Base Des Struct Mach 2022:1–20.
[39] Kalligeros C, Papalexis C, Vasileiou G, Tzouganakis P, Spitas C, Spitas V. Exploiting double-flank roll testing spur gear measurements to determine gear parameter deviations through numerical simulation of free-form meshing gears. Simulat Model Pract Theor 2023;126:102757.
[40] Wang J, Lv H. Modification and optimization of cycloidal gear tooth profile based on machining error compensation. Appl Sci 2023;13:2581.
[41] Tran TL, Pham AD, Ahn HJ. Lost motion analysis of one stage cycloid reducer considering tolerances. Int J Precis Eng Manuf 2016;17:1009–16.
[42] Sun X, Han L, Ma K, Li L, Wang J. Lost motion analysis of CBR reducer. Mech Mach Theor 2018;120:89–106.
[43] Xu H, Shi Z, Yu B, Wang H. Dynamic measurement of the lost motion of precision reducers in robots and the determination of optimal measurement speed. Journal of Advanced Mechanical Design, Systems, and Manufacturing 2019;13. JAMDSM0044.
[44] Li An, Deng. A new tooth profile modification method of cycloidal gears in precision reducers for robots. Appl Sci 2020;10:1266.
[45] Zhang C, Dong H, Dong B, Wang D. A bi-directional drive model for lost motion behavior of planetary gear train. Mech Mach Theor 2022;174:104885.
[46] Zhang C, Dong H, Wang D, Cheng Y. Tolerance synthesis for lost motion requirement of planetary gear train based on a mechanism model. Mech Mach Theor 2021;164:104405.
[47] Lin K-S, Chan K-Y, Lee J-J. Kinematic error analysis and tolerance allocation of cycloidal gear reducers. Mech Mach Theor 2018;124:73–91.
[48] Wu KY, Shih YP, Lee JJ. Kinematic error analysis of the rotor vector gear reducer with machining tolerances. J Braz Soc Mech Sci Eng 2020;42:1–16.
[49] Jiang ZH, Zhang XL, Liu JQ. A reliability evaluation method for RV reducer by combining multi-fidelity model and Bayesian updating technology. IOP Conf Ser Mater Sci Eng 2021;1043:052038.
[50] Chang S-L. Studies on epitrochoid gear for cycloid drives. J Mech 2003;19:271–8.
[51] Shin JH, Kwon SM. On the lobe profile design in a cycloid reducer using instant velocity center. Mech Mach Theor 2006;41:596–616.
[52] Sun X, Han L, Wang J. Design and transmission error analysis of CBR reducer. J Mech Des 2019;141:1.
[53] Krajnik P, Hashimoto F, Karpuschewski B, da Silva EJ, Axinte D. Grinding and fine finishing of future automotive powertrain components. CIRP Ann 2021;70: 589–610.
[54] Xincheng W, Luyang L, Jianfeng Y, Xuefeng Y, Huaming W, Xigang H. Tolerance allocation for precision cycloid reducer requrirement based on a BDTCA model. J Mech Des 2023;145:073401.
[55] Pham A-D, Ahn H-J. Rigid precision reducers for machining industrial robots. Int J Precis Eng Manuf 2021;22:1469–86.
[56] Yi J, Jin T, Zhou W, Deng Z. Theoretical and experimental analysis of temperature distribution during full tooth groove form grinding. J Manuf Process 2020;58: 101–15.
[57] Li Tianxing, Zhou Junxiang, Deng Xiaozhong, et al. A manufacturing error measurement methodology for a rotary vector reducer cycloidal gear based on a gear measuring center. Meas Sci Technol 2018;29:075006.
[58] Li X, Tang L, He H, Sun L. Design and load distribution analysis of the mismatched cycloid-pin gear pair in RV speed reducers. Machines 2022;10:672.
[59] Zhang C, Zhang J-r, Tang P. Analysis of tooth profile and accumulative pitch errors of end-toothed disc and its tolerance development. Mech Mach Theor 2018;122: 86–96.
[60] Xu LM, Fan F, Zhang Z, Chao XJ, Niu M. Fast on-machine profile characterization for grinding wheels and error compensation of wheel dressing. Precis Eng 2019;55: 417–25.
[61] Hu C, Geng G, Spanos PD. Stochastic dynamic load-sharing analysis of the closed differential planetary transmission gear system by the Monte Carlo method. Mech Mach Theor 2021;165:104420.
[62] Shields MD, Zhang J. The generalization of Latin hypercube sampling. Reliab Eng Syst Saf 2016;148:96–108.
[63] Hu Y, Li G, Zhu WD, Cui J. An elastic transmission error compensation method for rotary vector speed reducers based on error sensitivity analysis. Appl Sci 2020;10 (2):481.
[64] Clerc M. Discrete particle swarm optimization, illustrated by the traveling salesman problem. In: Onwubolu GC, Babu BV, editors. New optimization techniques in engineering. Berlin, Heidelberg: Springer Berlin Heidelberg; 2004. p. 219–39.